Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with Jump-Diffusion

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Abstract

Synthetic equity return paths underpin stress testing, portfolio validation, and training data for machine-learning pipelines, yet existing generators leave two gaps: single-asset stylized facts (heavy tails, regime persistence, volatility clustering) rarely appear together in one model, and the standard recipe for adding multi-asset factor structure on top of a per-asset generator distorts the marginal the generator was built to deliver. We closed both gaps with a two-stage construction. The first stage was a hybrid hidden Markov framework that discretized excess growth rates into states defined by quantiles of a fitted Laplace distribution and augmented regime switching with a Poisson jump-duration mechanism, enforcing realistic tail-state dwell times without iterative Baum-Welch optimization. The second stage was a closed-form variance-corrected single-index composer that rescaled the per-asset generator draw before insertion into the market factor, preserving the marginal variance while recovering the calibrated factor loading; a sign-preserving clip handled high-leverage assets and a coefficient-of-determination-driven branch handled index-tracking assets, with each composed series tagged by a persisted construction flag so downstream consumers could audit the chosen path. We evaluated both stages on a US equity and exchange-traded-fund universe calibrated against a decade of SPY history with a one-year hold-out, benchmarking against parametric, neural, semi-Markov, and residual-fit alternatives on distributional, tail, factor-recovery, and Value-at-Risk metrics. The hybrid marginals reproduced the stylized facts in-sample and out-of-sample, and the variance-corrected composer recovered the calibrated factor loading while preserving heavy-tailed character at tail-risk accuracy comparable to dedicated residual fits. We also documented one downstream-use caveat, an iid-residual diversification-return artifact under daily rebalancing, addressable via a uniform location-shift correction anchored to a documented long-run prior. All code and per-asset fitted marginals were released as part of the open-source `JumpHMM.jl` library.

Keywords: hidden Markov model; jump-diffusion; single-index model; synthetic financial data; variance correction; Value-at-Risk backtest; volatility clustering; heavy-tailed marginals.

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1 Introduction

Synthetic equity return paths are an input to many quantitative-finance workflows. Stress-testing a portfolio against plausible but unobserved market scenarios, validating a portfolio optimization algorithm under conditions the historical record does not contain, augmenting limited training sets for machine-learning pipelines, and backtesting an allocator across many independent draws all require synthetic data that behave statistically like real markets in the dimensions the downstream task cares about [1, 2]. Empirical equity excess growth rates exhibit three statistical regularities a faithful generative framework must reproduce together: a heavy-tailed (leptokurtic) marginal distribution, negligible linear autocorrelation of raw returns, and persistent volatility clustering, in which large changes are followed by large changes of either sign and small changes by small changes [3–5]. These regularities, called *stylized facts*, are the benchmark for synthetic financial data.

Two gaps in the literature leave practitioners without a drop-in solution. The first is the single-asset stylized-facts gap. Existing generative approaches each excel at one of the three regularities and miss others. Generalized autoregressive conditional heteroscedasticity (GARCH) models [6, 7] capture conditional variance dynamics but do not represent discrete regimes; stochastic-volatility and jump-diffusion models [8, 9] enrich tail behavior but lack volatility persistence after jumps; deep generative models [10–14] can learn complex distributions but struggle to reproduce volatility clustering [5]. Hidden Markov models (HMMs) offer a probabilistic regime-switching structure [15], yet standard HMMs revert too quickly to central regimes after extreme events and fail to dwell in high-volatility tail states long enough to reproduce the empirical volatility-clustering profile [16, 17]. The second gap is the multi-asset factor-structure gap. A per-asset generator that reproduces single-asset stylized facts on a given equity says nothing about how that equity should co-move with the market index or with other equities on the same day. The standard recipe for adding such cross-sectional structure is a linear factor model, either Sharpe's one-factor single-index model (SIM) [18] or a multi-factor variant such as Ross's arbitrage pricing theory [19] or the Fama-French construction [20]. Applied naively to a per-asset full-return generator, however, the SIM recipe inflates the marginal variance of the composed series quadratically in the factor loading, defeating the purpose of fitting a heavy-tailed generator in the first place. The opposite failure mode, drawing the residual from a Gaussian with variance taken from the real ordinary least squares fit, recovers the SIM coefficients exactly but erases the marginal structure entirely. Neither naive option satisfies both the stylized-facts requirement and the factor-structure requirement at once.

In this study, we introduce a two-stage construction that closes both gaps. The first stage is a hybrid hidden Markov framework that discretizes continuous excess growth rates into Laplace quantile-defined states and augments regime switching with a two-parameter Poisson jump-duration mechanism. The Laplace partition bypasses the iterative Baum-Welch algorithm [21, 22] via direct frequentist counting of regime transitions, scaling the framework to a 424-asset pipeline. The Poisson override forces the chain to dwell in high-volatility tail states for empirically realistic durations, closing the volatility-clustering gap that standard HMMs leave open. The per-state emissions are Student-t with $\nu = 5$ degrees of freedom, which matched the empirical kurtosis to within 1.3% on SPY where Normal emissions would underestimate by approximately 29%. The second stage is a variance-corrected single-index composition that takes the per-asset hybrid HMM marginal as input and rescales it by a closed-form scalar so the composed series inherits the generator's marginal variance while recovering the calibrated factor loading. A sign-preserving clip handles high-leverage assets where the market loading would otherwise exceed the generator variance, and a separate R^2-driven branch handles index-tracking assets where the calibrated coefficient of determination, rather than the generator marginal, is the quantity worth preserving. The same construction generalizes to multi-factor variants without changes to its structure.

We validated both stages on a 424-ticker US equity and ETF universe calibrated against the 2014–2024 SPY history, with 2025 held out for out-of-sample evaluation. The single-asset stage was assessed in-sample and out-of-sample against seven baselines (bootstrap resampling, Gaussian, Laplace, GARCH(1,1), a hidden semi-Markov model [17], a hidden Markov model without jumps, and a Gated Recurrent Unit [23] neural baseline) on Kolmogorov-Smirnov, Anderson-Darling, Wasserstein-1, and Hellinger pass rates for distributional fidelity, alongside the autocorrelation function of $|g|$ as a measure of volatility clustering. The multi-asset stage was assessed against five composers (the naive composition, a Gaussian single-index residual, a JumpHMM fit directly to OLS residuals, a Politis-Romano stationary bootstrap of the real residual history [24], and a per-ticker GARCH(1,1)-t

fit [7]) on factor-loading recovery, marginal fidelity, and a per-ticker one-day Value-at-Risk backtest assessed by Kupiec's unconditional-coverage test. We close with a documented downstream-use caveat, an iid-residual diversification-return artifact under daily rebalancing, addressed by a calibrated location-shift bias correction that preserves rank order, drawdown shape, and volatility envelope while shifting only the level.

2 Related Work

Equity-return generators have been studied since the geometric Brownian motion baseline of Black-Scholes-Merton [25] failed on the empirical regularities that drive risk and pricing. Mandelbrot [3] showed that price changes carry heavier tails than the Gaussian admits; Fama [26] reviewed the evidence that raw returns have near-zero linear autocorrelation; and Schwert [27] documented the slow decay of absolute-return autocorrelation now known as volatility clustering. The ARCH and GARCH conditional-variance models of Engle [6] and Bollerslev [7] reproduced the clustering but left light-tailed marginals unless fed a fat-tailed innovation, and represented neither discrete regimes nor sudden jumps [28]. Stochastic-volatility models [29, 8] treat volatility as a latent continuous process at the cost of heavy estimation. Jump-diffusion models [9, 30] inject Poisson events directly into the price path, producing heavy tails by construction and fitting equity markets where announcements and earnings surprises generate large discrete moves [31], but in standard form they offer no mechanism for the volatility persistence that follows a jump.

Hidden Markov models address this gap by embedding regime and jump behavior in a latent discrete chain. Hamilton [32] introduced regime-switching to economics in a two-state form, and later work applied it to volatility forecasting [33] and equity trading [34, 35]. Ryden et al. [16] found that Gaussian-emission HMMs matched heavy tails and the absence of return autocorrelation but failed on volatility clustering; Bulla and Bulla [17] recovered the clustering with hidden semi-Markov models that allowed realistic dwell times in high-volatility states. Both rely on the Baum-Welch EM algorithm [21], which is iterative and sensitive to initialization. Deep generative alternatives [12, 14, 13] learn the marginal directly but routinely miss volatility clustering, and recent surveys [1, 2, 5] agree that stylized-facts reproduction remains the primary quality criterion; a pretrained foundation-inference alternative for Markov jump processes [36] has not yet been evaluated on financial data. The hybrid HMM of the present paper picks the latent-state route: it discretizes returns through a Laplace quantile partition (so transitions can be counted directly without EM) and augments the chain with a Poisson jump-duration override that holds it in high-volatility states long enough to reproduce empirical clustering.

Univariate generators still need a recipe for multi-asset composition. The single-index model (SIM) of Sharpe [18] provides one: each asset's growth rate decomposes into a market-loaded term plus an idiosyncratic residual, reducing the parameter count relative to a full multivariate fit. The construction generalizes to multi-factor variants [19, 20] without changing its form. Practitioners face two residual specifications. The *full-return* specification fits a per-asset generator to the asset's own growth-rate history and plugs the draw in as the residual; it lets a generator calibrated for distributional interpretability or cached for other pipelines enter the factor stage unchanged, but it inflates the composed marginal variance quadratically in the factor loading. The *residual-fit* specification fits the generator to the OLS residual series and composes with no further correction, recovering the factor exactly at the cost of the per-asset generator's distributional richness. Mixture hidden Markov models [37] share latent state across assets but are demanding to estimate in high dimensions; copula approaches [38–40] separate marginals from dependence at the cost of copula-family selection. The variance-corrected composition developed here threads these constraints: a closed-form scalar lets a full-return generator enter the factor pipeline while preserving its marginal variance, with an auditable R^2-driven branch for index-tracking edge cases.

Validation of synthetic financial data combines two families of tests: goodness-of-fit tests for distributional fidelity [41–43] and autocorrelation diagnostics for temporal structure [4]. Recent work has emphasized downstream-task validation as a complement: a synthetic dataset that passes distributional tests can still fail at the task it was generated for. Booth and Fama [44] formalized one such concern: a daily-rebalanced allocator on iid-residual paths extracts a structural *diversification return* that is largely absent in real markets, where residual autocorrelation is small but positive [45]. We adopt the classical distributional metrics together with a downstream Value-at-Risk backtest scored by Kupiec's unconditional-coverage test, and we address the diversification-return artifact through a documented bias correction.

3 Method

3.1 Hybrid hidden Markov marginal generator

The dataset was 424 United States-listed equities and exchange-traded funds spanning ten years (2014–2024). To validate the hybrid hidden Markov framework on a single representative asset, we focused the headline analysis on the SPDR S&P 500 ETF (SPY), reserving the $2,766$ observations from January 3, 2014 to December 31, 2024 for in-sample fitting and the subsequent 249 trading days of 2025 for out-of-sample testing. The data consisted of daily open, high, low, and close prices and volume-weighted average price values. We computed the excess growth rate $G_{i,j}$ for ticker i between trading days $j-1$ and j from the equity price series $P_{i,j}$ as:

$$G_{i,j} \;\equiv\; \frac{1}{\Delta t}\,\ln\!\left(\frac{P_{i,j}}{P_{i,j-1}}\right) \;-\; r_f, \tag{1}$$

where $\Delta t = 1/252$ is the fraction of a trading year spanned by one trading day, and r_f is a constant continuously compounded risk-free rate derived from STRIPS bond yields. The excess growth rate has units of year^{-1} and is time-additive under continuous compounding.

We defined the hybrid hidden Markov model on this dataset as the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ [15], where the hidden state $S_t \in \mathcal{S} = \{1, \ldots, N\}$ represented market regimes, the observation space $\mathcal{O} \subset \mathbb{R}$ consisted of continuous excess growth rate values, $\mathbf{T}$ governed regime-to-regime transitions, $\mathbf{E}$ specified the state-conditional emission model, and $\bar{\pi}$ denoted the stationary distribution. States were defined by quantile boundaries of a fitted Laplace distribution: $Q_k = F_L^{-1}(k/N;\, \mu_L, b_L)$, where $F_L^{-1}(\cdot;\, \mu_L, b_L)$ is the inverse cumulative distribution function of the Laplace distribution with location μ_L and scale b_L (both estimated from the in-sample growth-rate history by maximum likelihood), with the endpoints fixed at $Q_0 = -\infty$ and $Q_N = +\infty$. The Laplace was chosen for its sharper peak (better matching the concentration of small price movements [3, 4, 46]) and its closed-form quantile function, which scaled to the 424-ticker pipeline without iterative optimization [22]. An observation was assigned to state k when $Q_{k-1} < G_t \le Q_k$. The within-state emission followed a location-scale Student-t distribution:

$$G_t \mid S_t = k \;\sim\; \mu_k + \sigma_k \cdot t_\nu, \qquad \nu = 5, \tag{2}$$

where t_ν denotes a standard Student-t random variable with ν degrees of freedom, and μ_k, σ_k are state-conditional location and scale parameters defined below. The value $\nu = 5$ was selected from a sensitivity analysis over $\nu \in \{3, 4, 5, 6, 7, 8, 10, 15, 30, \infty\}$ as the value that minimized the kurtosis gap without degrading distributional fidelity ($\nu = \infty$ recovers the Normal baseline).

With the state space and emission family fixed, we estimated the transition matrix $\mathbf{T}$ through direct frequentist counting: the discrete state assignments allowed straightforward counting of observed transitions between regimes, which avoided the initialization sensitivity of expectation-maximization [22]. A consequence of direct counting was that empirical transition matrices often had low probabilities for exiting central states into tail states, which caused models to lose the effect of volatility clustering too quickly [17]. We corrected this by adding two jump parameters, ϵ representing the probability of a jump event and λ representing the mean jump-event duration, along with a tail-state selection rule. Tail states were defined as $\mathcal{S}_- = \{1, \ldots, N_{\mathrm{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\mathrm{tail}} + 1, \ldots, N\}$, where N_{tail} was user-configurable. When a jump was triggered, the jump duration K was sampled from a Poisson distribution $K \sim \mathrm{Poisson}(\lambda)$, which is standard for modeling independent events in fixed intervals [47]. During these K steps, the standard Markovian transitions were overridden to target tail states, with a user-configurable bias p_{neg} (default 0.52) that favored negative-tail states to reproduce gain/loss asymmetry (Figure 1). Given the state assignments and transition matrix, the remaining model components were estimated directly from the data: the emission parameters μ_k and σ_k were the sample mean and standard deviation of observations assigned to state k, and the empirical transition matrix exhibited a dominant near-diagonal band reflecting short-range regime persistence (Online Appendix E). The stationary distribution $\bar{\pi}$ satisfied $\bar{\pi} = \bar{\pi}\mathbf{T}$ and was estimated by matrix powering ($\mathbf{T}^{50}$) [48, 49]. An initial state was drawn from $S_0 \sim \bar{\pi}$; at each subsequent time step t the chain either transitioned according to the empirical row $\mathbf{T}_{S_{t-1},\cdot}$ (probability $1-\epsilon$) or

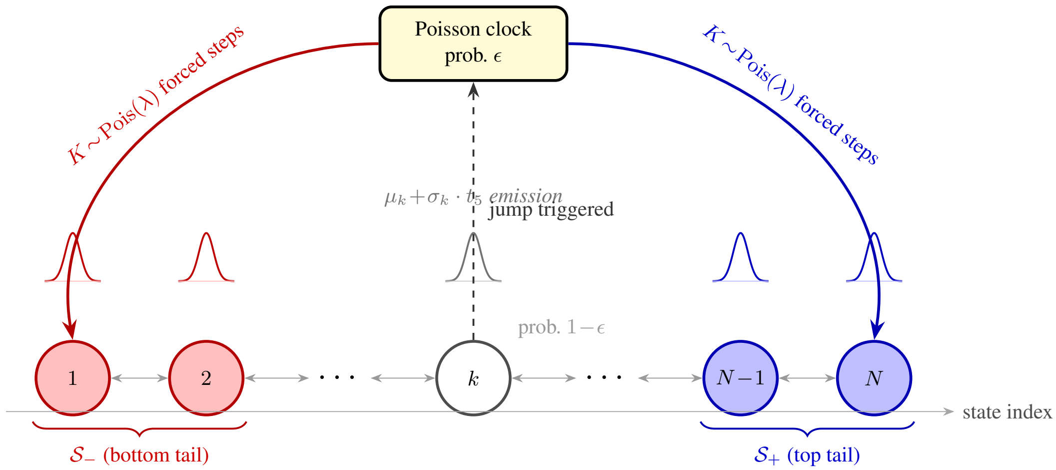

Figure 1: **Architecture of the hybrid hidden Markov model with jump-diffusion (HMM-WJ).** At each step the chain transitions according to the empirical transition matrix $\mathbf{T}$ (probability $1 - \epsilon$, thin bidirectional arrows) or enters a Poisson jump episode (probability ϵ, dashed upward arrow to the Poisson clock). During a jump episode the active state is forced to the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) for $K \sim \text{Poisson}(\lambda)$ consecutive steps before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional location-scale Student-t ($\nu = 5$) emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

entered a Poisson jump episode (probability ϵ) that forced the state into $\mathcal{S}_-$ or $\mathcal{S}_+$ for K consecutive steps before normal transitions resumed. In both cases an observation was sampled from the emission distribution $G_t \sim \mathbf{E}_{S_t}$. The full simulation pseudocode is given in Online Appendix D.

With the simulator specified, we calibrated the two jump parameters, the jump probability ϵ and mean jump duration λ [9, 30], via a multi-objective grid search that minimized the discrepancy between historical and simulated market signatures. The objective targeted two stylized facts [4, 16]: volatility clustering, measured as the squared error between the observed and simulated autocorrelation function (ACF) of absolute excess growth rates [27] up to a lag of $L = 252$ trading days (approximately one trading year), and leptokurtosis, captured by a penalty term on the global kurtosis [3]:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^{L} \left(\text{ACF}_{\text{obs}}(\tau) - \overline{\text{ACF}}_{\text{sim}}(\tau)\right)^2 + w_K \left(K_{\text{obs}} - \overline{K}_{\text{sim}}\right)^2, \tag{3}$$

where $\text{ACF}_{\text{obs}}(\tau)$ and K_{obs} are the empirical absolute-growth autocorrelation at lag τ and the global kurtosis, respectively, and the simulated counterparts $\overline{\text{ACF}}_{\text{sim}}(\tau)$ and $\overline{K}_{\text{sim}}$ were averaged across 200 independent synthetic paths of 2,766 trading days per grid point [47]. We set the kurtosis-penalty weight to $w_K = 0.20$ to balance tail behavior against the temporal ACF term. The grid explored ϵ from 10^{-4} to 2.5×10^{-2} and λ from 10 to 160, and the full pseudocode is given in Online Appendix D. The framework above produces a per-ticker marginal that reproduces heavy tails, regime persistence, and volatility clustering (validated empirically below); multi-asset workflows that consume these paths need a cross-sectional structure on top, which the next subsection supplies via a closed-form composition rule.

3.2 Variance-corrected single-index composition

As in the preceding subsection, we work in continuously compounded annualized excess growth rates: $g(t) = \Delta t^{-1} \ln[p(t)/p(t-1)] - r_f$, where $p(t)$ is the closing price on trading day t, r_f is a constant continuously compounded risk-free rate, and $\Delta t = 1/252$ is the fraction of a trading year spanned by one trading day. Annualized growth rates keep variances and covariances in the units customary in risk reporting, and they compose linearly under aggregation in time. Throughout this subsection, the sample mean, sample variance, and sample covariance of any length-T real vector $x \in \mathbb{R}^T$ (and any two such vectors x, y) are written $\bar{x}$, $\text{Var}(x)$, and $\text{Cov}(x, y)$; all three are computed over the T time steps of the path in the usual way. The construction composes a synthetic per-asset growth-rate path g_i under the single-index model (SIM):

$$g_i(t) \;=\; \alpha_i \;+\; \beta_i\, g_m(t) \;+\; \varepsilon_i(t), \qquad t = 1, \ldots, T. \tag{4}$$

Here, $\tilde{g}_i$ is a length-T growth-rate path produced by the per-asset generator (a tilde marks the path as a generator output, to distinguish it from the real or composed paths g_i and g_m, which carry no tilde). The residual ε_i in the SIM is a scaled copy of $\tilde{g}_i$: $\varepsilon_i(t) = s_i\, \tilde{g}_i(t)$, with $s_i \in (0, 1]$ a scalar chosen so the composed path inherits one of two per-asset marginal targets while recovering the calibrated SIM coefficients (α_i, β_i) in ordinary least squares (OLS). The next paragraph defines the inputs formally; the rest of the subsection derives s_i and a sign-preserving clip on β_i that handles high-leverage assets.

The construction takes three inputs. The first input is the *market growth-rate path*: a length-T real-valued vector $g_m = (g_m(1), \ldots, g_m(T)) \in \mathbb{R}^T$ giving the market's excess growth rate on each of T trading days, with sample mean μ_m and sample variance $\sigma_m^2 = \mathrm{Var}(g_m)$. In the experiments below, g_m is the SPY total-return excess growth-rate series; the construction does not depend on this choice, and g_m can equally be a single-name proxy, a synthetic index, or a model-generated path. The second input is, for each of N assets indexed by $i = 1, \ldots, N$, a per-asset *generator growth-rate path*: a length-T real-valued vector $\tilde{g}_i = (\tilde{g}_i(1), \ldots, \tilde{g}_i(T)) \in \mathbb{R}^T$ produced by a probabilistic model of the asset's growth-rate dynamics. In this paper the generator is the hybrid hidden Markov model developed above, whose single-time-step marginal is a mixture of state-conditional Student-t ($\nu = 5$) emissions over the $N = 100$ Laplace-quantile states, and whose path-level structure adds regime persistence and occasional Poisson jump episodes (Eq. (2), Fig. 1). The SIM composition below is generator-agnostic, however: it uses only the values of $\tilde{g}_i$ and its sample variance $\sigma^2_{\mathrm{gen},i} = \mathrm{Var}(\tilde{g}_i)$, and assumes no parametric form for the marginal distribution. Two operating conditions are imposed on each generator path. First, $\tilde{g}_i$ has sample mean zero, so any drift carried by the asset is absorbed by the SIM intercept α_i rather than by the generator. Second, $\tilde{g}_i$ is approximately uncorrelated with the market path, $\mathrm{Cov}(\tilde{g}_i, g_m) \approx 0$, which is automatic when the generator is fit and sampled in isolation, and which we verify empirically across the universe (Fig. S1). The third input is a per-asset *calibration triple* $(\alpha_i, \beta_i, R^2_{i,\mathrm{real}})$ obtained by OLS of the asset's real excess growth-rate history on g_m: α_i and β_i are the OLS intercept and slope, and $R^2_{i,\mathrm{real}}$ is the coefficient of determination. These are the targets the synthetic construction must recover; $R^2_{i,\mathrm{real}}$ also selects which of two branches (variance-preserving or R^2-preserving) is used by the construction below.

Given these inputs, we want the synthetic composition $g_i = \alpha_i + \beta_i^{\mathrm{eff}}\, g_m + \varepsilon_i$, with effective slope β_i^{eff} (equal to β_i on the unclipped branch and the attenuated value of Eq. (10) otherwise), to satisfy three properties in the large-T limit. The first is *SIM recovery*: ordinary least squares of g_i on g_m returns the calibrated intercept and slope, $\hat{\alpha}_i = \alpha_i$ and $\hat{\beta}_i = \beta_i^{\mathrm{eff}}$; on the unclipped branch, $\beta_i^{\mathrm{eff}} = \beta_i$ and the OLS output reproduces the calibration exactly, while on the clipped branch the attenuation is persisted alongside the output so it is auditable. The second is *marginal fidelity*: each asset is assigned one of two per-asset targets based on its calibrated R^2, not both at once. Assets whose generator marginal (heavy tails, regime structure, jumps) is the quantity of interest receive the variance-preserving target $\mathrm{Var}(g_i) = \sigma^2_{\mathrm{gen},i}$; assets that track the market closely receive the R^2-preserving target $R^2 = R^2_{i,\mathrm{real}}$, measured by the same OLS regression as above. The branch chosen for each asset is recorded as a construction flag so downstream consumers can audit which target was pursued. The third is *cross-sectional dependence*: any joint dependence structure applied to the residuals $\{\varepsilon_i\}_{i=1}^N$ passes through unchanged to the composed series $\{g_i\}_{i=1}^N$, automatic given the linearity of Eq. (4) in ε_i; the joint structure itself (empirical copula, factor copula, t-copula) is supplied as a separate design choice, and we report cross-sectional fidelity for the empirical-copula choice in the results below. The naive composition $g_i = \alpha_i + \beta_i\, g_m + \tilde{g}_i$ satisfies SIM recovery and cross-sectional dependence but violates marginal fidelity: its marginal variance is $\sigma^2_{\mathrm{gen},i} + \beta_i^2 \sigma_m^2$, an inflation that grows quadratically in β_i. The rest of this subsection fixes this inflation with a closed-form scalar correction, handles the degenerate high-β case with a sign-preserving clip, treats index-tracking assets through an R^2-preserving branch, and ties the pieces together in an algorithm that states the resulting properties of the composed series.

To fix the inflation, form the candidate residual $\varepsilon_i = s_i\, \tilde{g}_i$ for some scalar $s_i \in (0, 1]$, and require that the composed series have the same variance as the generator path:

$$\mathrm{Var}(g_i) \;=\; \mathrm{Var}(\alpha_i + \beta_i\, g_m + \varepsilon_i) \;=\; \mathrm{Var}(\beta_i\, g_m + \varepsilon_i) \;=\; \sigma^2_{\mathrm{gen},i}, \tag{5}$$

where α_i drops out because it is a constant offset and adds nothing to the variance. Expanding the variance of the sum gives:

$$\mathrm{Var}(\beta_i\, g_m + \varepsilon_i) \;=\; \beta_i^2\; \mathrm{Var}(g_m) \;+\; 2\beta_i\; \mathrm{Cov}(g_m,\, \varepsilon_i) \;+\; \mathrm{Var}(\varepsilon_i). \tag{6}$$

Substituting $\varepsilon_i = s_i\, \tilde{g}_i$ in (6) yields $\mathrm{Var}(\varepsilon_i) = s_i^2\, \sigma_{\mathrm{gen},i}^2$ and $\mathrm{Cov}(g_m,\, \varepsilon_i) = s_i\; \mathrm{Cov}(g_m,\, \tilde{g}_i)$, so under the independence assumption $\mathrm{Cov}(\tilde{g}_i,\, g_m) \approx 0$ the middle (cross) term vanishes and the variance-target condition (5) reduces to:

$$\beta_i^2\, \sigma_m^2 \;+\; s_i^2\, \sigma_{\mathrm{gen},i}^2 \;=\; \sigma_{\mathrm{gen},i}^2, \tag{7}$$

which gives:

$$\boxed{s_i^2 \;=\; 1 \;-\; \frac{\beta_i^2\, \sigma_m^2}{\sigma_{\mathrm{gen},i}^2}} \tag{8}$$

Define the *market loading ratio*:

$$\rho_i \;\equiv\; \frac{\beta_i^2\, \sigma_m^2}{\sigma_{\mathrm{gen},i}^2}, \tag{9}$$

the share of the generator variance consumed by the market component; then $s_i^2 = 1 - \rho_i$, valid whenever $\rho_i < 1$. The naive composition $g_i = \alpha_i + \beta_i g_m + \tilde{g}_i$ corresponds to $s_i = 1$ and inflates $\mathrm{Var}(g_i)$ by exactly $\beta_i^2 \sigma_m^2$. Equation (8) breaks down when $\rho_i \geq 1$: the market loading alone would account for more than the entire generator variance, leaving no room for idiosyncratic noise, a case that arises for high-β_i assets paired with low-volatility generators or whenever the synthetic market g_m is more volatile than the market on which β_i was calibrated. To handle it, impose a floor $f \in (0, 1)$ on the idiosyncratic share by requiring $s_i^2 \geq f$, and clip β_i downward whenever the floor would be violated. Writing the clip threshold as $\rho_i^{\max} = 1 - f$, the piecewise rule is:

$$\beta_i^{\mathrm{eff}} \;=\; \begin{cases} \beta_i & \text{if } \rho_i \leq \rho_i^{\max} \\ \mathrm{sign}(\beta_i)\, \sqrt{(1-f)\, \dfrac{\sigma_{\mathrm{gen},i}^2}{\sigma_m^2}} & \text{if } \rho_i > \rho_i^{\max} \end{cases} \qquad s_i^2 \;=\; \begin{cases} 1 - \rho_i & \text{if } \rho_i \leq \rho_i^{\max} \\ f & \text{if } \rho_i > \rho_i^{\max} \end{cases} \tag{10}$$

Sign preservation in (10) keeps defensive (negative-β) assets defensive even when clipped. The floor f is a tunable knob; in the experiments below we use $f = 0.10$, which guarantees that at least 10% of each asset's variance comes from idiosyncratic noise. When β_i is clipped we log a warning and persist both β_i and β_i^{eff} so downstream consumers can audit the attenuation.

The variance-preserving construction above targets $\mathrm{Var}(g_i) = \sigma_{\mathrm{gen},i}^2$, which is the right target when the generator marginal carries information one wants to keep. For some assets the generator marginal is not really the quantity of interest; the SIM relationship to the market is. Index ETFs are the standard example: SPY against itself has $\beta = 1$, $R^2 = 1$, $\sigma_\varepsilon = 0$ exactly; QQQ against SPY has $R^2 \approx 0.84$, and SPYG against SPY has $R^2 \approx 0.86$. For these tickers we require $\hat{\beta}$ and R^2 to recover their calibrated values, neither inflated to one by dropping ε nor implied by unrelated marginal-variance bookkeeping. We branch on the real-data calibration $R_{i,\mathrm{real}}^2$: pick a threshold R_{preserve}^2 that separates index-tracking assets from genuine single stocks; in our universe the cumulative distribution sorts itself cleanly, with index ETFs above 0.80 and the most market-like single names below 0.65 (Fig. S3), so $R_{\mathrm{preserve}}^2 = 0.80$ captures the trackers without sweeping in any genuinely idiosyncratic name. For any asset with $R_{i,\mathrm{real}}^2 \geq R_{\mathrm{preserve}}^2$, the population R^2 identity for (4) is:

$$R^2 \;=\; \frac{\beta_i^2\, \sigma_m^2}{\beta_i^2\, \sigma_m^2 + \sigma_{\varepsilon,i}^2}. \tag{11}$$

This inverts to give the residual variance that hits the calibrated R^2:

$$\boxed{\sigma_{\varepsilon,\mathrm{target}}^2 \;=\; \beta_i^2\, \sigma_m^2 \cdot \frac{1 - R_{i,\mathrm{real}}^2}{R_{i,\mathrm{real}}^2}} \tag{12}$$

Draw $\tilde{g}_i$ from the ticker's generator and rescale:

$$\varepsilon_i \;=\; \sqrt{\frac{\sigma_{\varepsilon,\mathrm{target}}^2}{\sigma_{\mathrm{gen},i}^2}}\, \tilde{g}_i, \tag{13}$$

Algorithm 1 Hybrid SIM composition with variance correction.

Require: Market growth-rate path $g_m \in \mathbb{R}^T$ with sample variance σ_m^2; per-asset generator growth-rate paths $\tilde{g}_i \in \mathbb{R}^T$ for $i = 1, \ldots, N$; calibrated SIM parameters $(\alpha_i, \beta_i, R^2_{i,\text{real}})$; floor $f \in (0, 1)$; threshold $R^2_{\text{preserve}} \in (0, 1)$.

Ensure: Composed paths $g_i \in \mathbb{R}^T$ and per-asset construction flags.

1: **for** each asset $i = 1, \ldots, N$ **do**
2: Compute generator variance $\sigma^2_{\text{gen},i} \leftarrow \mathrm{Var}(\tilde{g}_i)$.
3: **if** $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ **then**
4: $\sigma^2_{\varepsilon,\text{target}} \leftarrow \beta_i^2 \sigma_m^2 (1 - R^2_{i,\text{real}})/R^2_{i,\text{real}}$ {Eq. (12)}
5: $\varepsilon_i \leftarrow \sqrt{\sigma^2_{\varepsilon,\text{target}}/\sigma^2_{\text{gen},i}}\, \tilde{g}_i$
6: $\beta_i^{\text{eff}} \leftarrow \beta_i$; branch $\leftarrow$ R^2-preserving
7: **else**
8: $\rho_i \leftarrow \beta_i^2 \sigma_m^2 / \sigma^2_{\text{gen},i}$ {Eq. (9)}
9: **if** $\rho_i \leq 1 - f$ **then**
10: $s_i^2 \leftarrow 1 - \rho_i$; $\beta_i^{\text{eff}} \leftarrow \beta_i$; branch $\leftarrow$ variance-preserving
11: **else**
12: $s_i^2 \leftarrow f$; $\beta_i^{\text{eff}} \leftarrow \mathrm{sign}(\beta_i)\sqrt{(1-f)\, \sigma^2_{\text{gen},i}/\sigma_m^2}$; branch $\leftarrow$ clipped variance-preserving
13: **end if**
14: $\varepsilon_i \leftarrow s_i\, \tilde{g}_i$
15: **end if**
16: *(Optional)* apply cross-sectional rank reorder to ε_i.
17: $g_i \leftarrow \alpha_i + \beta_i^{\text{eff}}\, g_m + \varepsilon_i$ {Eq. (4)}
18: **end for**

then copula-reorder and compose as in (4). The construction recovers both calibrated quantities by design: $\hat{\beta} \to \beta_i$ in the large-T limit because ε_i is uncorrelated with g_m after the rank reorder, and $R^2 \to R^2_{i,\text{real}}$ by construction. Setting $R^2_{i,\text{real}} = 1$ in Eq. (12) gives $\sigma^2_{\varepsilon,\text{target}} = 0$, so the SPY-against-itself identity $g_i = \alpha_i + \beta_i g_m$ falls out deterministically without special-casing; conversely, when the rescale factor in Eq. (13) exceeds one (which happens when the synthetic market g_m has lower variance than the calibration market) the generator's tails are stretched to fit a target larger than the original draw, and we detect and flag the case so the loss of marginal fidelity is auditable.

The pieces combine into a single procedure (Algorithm 1). For each asset the calibrated $R^2_{i,\text{real}}$ selects a branch: assets with $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ go through the R^2-preserving construction of Eq. (12), and the rest go through the variance-preserving construction of Eq. (8) with the clip of Eq. (10) engaged when the market loading ratio ρ_i would otherwise exceed $1 - f$. Each asset is labelled with the branch that produced it (R^2-preserving, variance-preserving, or clipped variance-preserving), recorded alongside the output so the path taken per asset can be recovered without external bookkeeping. The optional rank reorder in line 14 accepts any joint dependence structure (empirical copula, t-copula, factor copula); reordering preserves the marginal variance of ε_i, so the variance algebra of Eq. (8) is unaffected and whatever cross-sectional structure is applied to ε_i passes through to g_i unchanged.

By construction the composed series g_i satisfies the three preservation criteria above (SIM recovery, marginal fidelity, cross-sectional dependence) in the large-T limit. OLS of g_i on g_m recovers $\hat{\alpha}_i \to \alpha_i$ and $\hat{\beta}_i \to \beta_i^{\text{eff}}$, with $\beta_i^{\text{eff}} = \beta_i$ unclipped and β_i^{eff} persisted alongside β_i when the clip triggers. Marginal variance matches $\sigma^2_{\text{gen},i}$ on the variance-preserving branch, either exactly, $\mathrm{Var}(g_i) = \beta_i^2 \sigma_m^2 + s_i^2 \sigma^2_{\text{gen},i} = \sigma^2_{\text{gen},i}$, or by the choice of β_i^{eff} in (10) under clipping; on the R^2-preserving branch the marginal variance instead matches $\beta_i^2 \sigma_m^2 / R^2_{i,\text{real}}$, the value implied by the calibrated $(\beta_i, R^2_{i,\text{real}})$ rather than by the generator marginal. Tail behavior on the variance-preserving branch follows from the floor: because at least $f\, \sigma^2_{\text{gen},i}$ of the variance is reserved for ε_i, the tails inherited from $\tilde{g}_i$ dominate whenever $(\beta_i^{\text{eff}})^2 \sigma_m^2$ is small relative to those tails, while as β_i grows the marginal of g_i tightens toward Gaussian along the market-driven axis, mildly for low-β assets and noticeably at high β. The construction injects *contemporaneous* market dependence only; lagged dependence (lead-lag effects, market-driven volatility clustering) requires a richer construction, which we return to in the discussion. The pipeline is modest in compute: fitting the 423 per-asset JumpHMM marginals runs once offline against the 2014–2024 price histories and is cached, while GARCH(1,1)-t residual paths are simulated per (ticker, replication) at scoring time rather than cached, so each run consumes a fresh draw. The scoring loop across $423 \times 100 = 42{,}300$ per-ticker replications per composer runs in minutes on a single laptop, and the full six-composer sweep reported below completes within an hour on the same hardware.

4 Results

We validated the two-stage construction in two passes. The first pass asked whether the hybrid hidden Markov framework reproduced single-asset stylized facts on real SPY data, in-sample and out-of-sample, against six baseline generators. The second pass asked whether the variance-corrected single-index composer preserved those validated marginals when combined with a market factor across a 424-ticker universe, against five alternative composers, and whether the resulting paths supported a downstream Value-at-Risk backtest at industry-standard coverage levels. Descriptive statistics confirmed that all three stylized facts were present in both the in-sample (2014–2024, $T = 2{,}766$) and out-of-sample (2025, $T = 249$) windows (Table S1): excess kurtosis of 7.7 (IS) and 6.9 (OoS) with Jarque-Bera rejection ($p < 0.001$), persistent volatility clustering (Ljung-Box on $|G_t|$ rejected, $p < 0.001$), and near-zero linear autocorrelation in raw returns [26].

With the stylized facts confirmed in both evaluation windows, we calibrated the jump mechanism. A multi-objective grid search over $\epsilon \in [10^{-4},\, 2.5 \times 10^{-2}]$ and $\lambda \in [10,\, 160]$, with 200 simulated paths per grid point, identified the optimal values $\epsilon^* = 10^{-4}$ and $\lambda^* = 100$ (Online Appendix D). The optimal $\lambda^* = 100$ was an interior point of the grid, confirming that the mean jump duration was identifiable from data; $\epsilon^* = 10^{-4}$ fell at the lower boundary of the ϵ grid, indicating that the data consistently preferred rarer tail-entry events over the range explored. At these optimal values, the ACF of $|G_t|$ closely tracked the observed SPY ACF across lags 1–252, confirming that the jump mechanism partially reproduced the empirical ARCH effect. The primary analysis used $N = 100$ states, but a sensitivity study over $N \in \{30, 60, 90, 100, 150, 200\}$ confirmed robustness: KS and AD pass rates remained stable for both HMM-NJ (without jumps) and HMM-WJ (with jumps) across this range, with every state receiving adequate statistical support (Online Appendix F). At $N = 350$, certain states were never visited in the historical record, establishing a practical upper bound on state resolution.

With the model calibrated, we benchmarked HMM-WJ against seven alternative generators on $1{,}000$ synthetic paths of $2{,}766$ trading days each (Table 1, Figure 2). Bootstrap resampling set the non-parametric ceiling (100% KS, 100% coverage), while the Gaussian i.i.d. baseline (0% KS) confirmed that thin-tailed models were inadequate. Between these anchors, every model excelled on one quality dimension but failed on another. The GARCH(1,1) baseline achieved the best temporal fidelity (ACF-MAE 0.031) but a poor distributional fit (5.5% KS), indicating a systematic shape mismatch despite its volatility clustering properties. The Gated Recurrent Unit (GRU) [23] closed 83% of the ACF-MAE gap but produced a 0.6% KS pass rate. The hidden semi-Markov model (HSMM) [17] occupied a middle ground (82% KS) but its coarse 8-state partition could not capture the full growth-rate distribution, as reflected by a 38% kurtosis gap and only 68.7% quantile coverage. No single model captured both quality dimensions. Both HMM variants achieved the highest distributional fidelity among all parametric models, with HMM-WJ's pass rates 2–8 percentage points below HMM-NJ's, a cost of the jump mechanism occasionally overweighting tail states. The Student-t ($\nu = 5$) emissions closely matched the observed kurtosis (within 1.3% for HMM-WJ), a substantial improvement over Normal emissions which underestimated kurtosis by approximately 29%. Negative skewness (-0.75 observed) was also well reproduced by both HMM variants, arising from the asymmetric state-occupancy distribution inherited from the Laplace partition rather than any explicit skewness parameter; symmetric-innovation models (GARCH, Gaussian, Laplace) all produced near-zero skewness by construction. On the temporal dimension, HMM-NJ scored 0.059 on ACF-MAE, closing only 3% of the gap between the i.i.d. floor (0.060) and the GARCH ceiling (0.031); HMM-WJ scored 0.052, closing 28% of the gap and the only non-GARCH non-neural model to reduce ACF-MAE meaningfully below the i.i.d. floor, because approximately 24% of HMM-WJ paths contained at least one Poisson jump event that sustained $\mathrm{ACF}(|G_t|)$ decay across all lags while the remaining 76% behaved identically to HMM-NJ. Out-of-sample on the held-out 249

Table 1: **In-sample and out-of-sample model comparison for SPY** (1,000 simulated paths, significance level $\alpha = 0.05$). Bootstrap: i.i.d. resample of training data. Gaussian/Laplace: i.i.d. draws from MLE-fitted distributions. GARCH: GARCH(1,1) estimated by quasi-maximum likelihood. GRU: 2-layer gated recurrent unit [23] with Gaussian output head. HSMM: hidden semi-Markov model with $K = 8$ Laplace quantile states, negative-binomial dwell times, and Student-t ($\nu = 5$) emissions [17]. HMM-NJ: hidden Markov model ($N = 100$ states) without jumps. HMM-WJ: hybrid model with Poisson jump-duration extension. ACF-MAE: mean absolute error of the ACF of $|G_t|$ over lags 1–252. Wasserstein-1 and Hellinger distance defined in Online Appendix C. Coverage: fraction of 99 empirical quantiles (1st–99th) falling within the [5th, 95th] percentile envelope of the corresponding synthetic quantiles. Standard errors in parentheses: binomial SE for pass rates, $\mathrm{std}/\sqrt{n}$ for kurtosis, Wasserstein-1, and Hellinger; bootstrap SE ($B = 500$) for ACF-MAE and coverage. Best value per row in bold.

Metric	**Bootstrap**	**Gaussian**	**Laplace**	**GARCH(1,1)**	**GRU**	**HSMM**	**HMM-NJ**	**HMM-WJ**
In-sample: 2,766 *trading days* (2014–2024)								
KS pass rate (%) ↑	100.0 (<0.1)	0.0 (<0.1)	44.0 (1.6)	5.5 (0.7)	0.6 (0.2)	82.0 (1.2)	**99.7** (0.2)	97.6 (0.5)
AD pass rate (%) ↑	99.7 (0.2)	0.0 (<0.1)	43.3 (1.6)	1.9 (0.4)	0.2 (0.1)	42.5 (1.6)	**99.1** (0.3)	91.3 (0.9)
Excess kurtosis (observed)	7.715							
Excess kurtosis (simulated) ≈	7.6 (0.08)	−0.0 (<0.01)	3.0 (0.02)	**8.2** (0.37)	5.7 (0.16)	4.8 (0.08)	8.1 (0.14)	7.6 (0.14)
ACF-MAE ↓	0.060 (<0.001)	0.060 (<0.001)	0.060 (<0.001)	**0.031** (0.001)	0.036 (<0.001)	0.059 (<0.001)	0.059 (<0.001)	0.052 (<0.001)
Coverage (%) ↑	100.0 (<0.1)	13.1 (0.1)	37.4 (1.5)	29.3 (1.4)	17.2 (0.5)	68.7 (1.1)	**100.0** (<0.1)	**100.0** (<0.1)
Wasserstein-1 ↓	0.062 (0.001)	0.399 (0.001)	0.138 (0.001)	0.304 (0.006)	0.421 (0.003)	0.176 (0.001)	**0.081** (0.001)	0.101 (0.001)
Hellinger dist ↓	0.042 (<0.001)	0.148 (<0.001)	**0.072** (<0.001)	0.100 (0.001)	0.134 (0.001)	0.113 (<0.001)	0.073 (<0.001)	0.075 (<0.001)
Out-of-sample: 249 *trading days* (2025)								
KS pass rate (%) ↑	97.4 (0.5)	62.2 (1.5)	88.0 (1.0)	80.3 (1.3)	71.8 (1.4)	96.2 (0.6)	**96.7** (0.6)	94.4 (0.7)
AD pass rate (%) ↑	98.5 (0.4)	46.3 (1.6)	92.9 (0.8)	72.0 (1.4)	44.8 (1.6)	**96.7** (0.6)	**96.7** (0.6)	95.1 (0.7)
Excess kurtosis (observed)	6.867							
Excess kurtosis (simulated) ≈	6.0 (0.18)	−0.0 (<0.01)	2.7 (0.05)	1.6 (0.07)	2.9 (0.10)	4.1 (0.13)	**6.4** (0.22)	6.1 (0.13)
ACF-MAE ↓	0.043 (<0.001)	0.043 (<0.001)	0.043 (<0.001)	**0.026** (<0.001)	0.033 (<0.001)	0.042 (<0.001)	0.041 (<0.001)	0.039 (<0.001)
Coverage (%) ↑	100.0 (<0.1)	36.4 (1.9)	74.7 (1.0)	96.0 (1.3)	77.8 (2.9)	**100.0** (0.2)	**100.0** (<0.1)	**100.0** (<0.1)
Wasserstein-1 ↓	0.232 (0.002)	0.452 (0.002)	0.263 (0.002)	0.507 (0.018)	0.488 (0.005)	0.287 (0.002)	**0.258** (0.002)	0.282 (0.006)
Hellinger dist ↓	0.205 (0.001)	0.235 (0.001)	0.211 (0.001)	0.232 (0.001)	0.240 (0.001)	0.239 (0.001)	**0.207** (0.001)	0.210 (0.001)
Parameters estimated	0	2	2	3	37,954	18	2	4

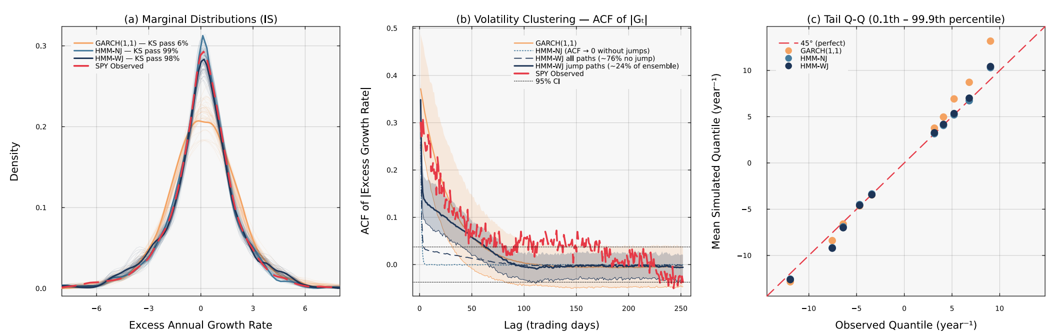

Figure 2: **Head-to-head in-sample model comparison for SPY** ($N = 100$, 1,000 simulated paths). *Panel (a):* marginal density of excess growth rates with in-sample KS pass rates annotated. GARCH(1,1) fails the two-sample KS test on 95% of paths, indicating a systematic shape mismatch with the empirical distribution; both HMM variants pass at rates $\geq 97\%$. *Panel (b):* autocorrelation function of $|G_t|$ at lags 1–252 (shaded bands: 10th–90th percentile across paths). HMM-NJ is structurally incapable of producing persistent volatility clustering; HMM-WJ generates a mixture of two path families, approximately 76% behaving like HMM-NJ and the remaining $\sim$24% containing at least one Poisson jump that sustains $\mathrm{ACF}(|G_t|)$ decay across all lags. *Panel (c):* tail Q-Q plot at the 0.1st–99.9th percentile region; HMM-WJ provides the closest mean quantile match in the extreme tails.

trading days of 2025 the same ranking held (Table 1, Figure S4): HMM-NJ achieved 96.7% KS and HMM-WJ achieved 94.4% KS, with simulated kurtosis of 6.4 and 6.1 against an observed 6.9, and HMM-WJ remained the only model to improve meaningfully on both dimensions simultaneously. GARCH's W_1 deteriorated by 67% from in-sample to out-of-sample, exposing a tail misspecification that binary pass rates obscured. Cross-asset generalization to NVDA, JNJ, and JPM confirmed that the jump mechanism automatically adapted its duration λ^* to each asset's clustering intensity (Online Appendix G).

Having validated the per-asset marginals on single-asset stylized facts in both windows, we turned to the multi-asset composition pipeline and asked whether those validated marginals plugged into a multi-asset construction without breaking either target. We evaluated six composers on a fixed set of per-asset JumpHMM marginals calibrated against the real SPY growth-rate path. This isolated the effect of the composition rule from the effect of market-path randomness: all composers consumed the same real SPY history and the same per-ticker innovation draw, so metric differences across composers reflected the composition rule alone. The ticker universe was a 424-asset US equity/ETF set drawn from Polygon.io daily-close histories, filtered to tickers with a complete in-sample trading history matching the reference length of AAPL ($T = 2{,}767$ daily closes covering January 2014 through December 2024); filtering on the common length preserved 424 tickers, of which the SPY total-return growth-rate series was taken as the market path g_m and the remaining 423 tickers as non-market assets. For each non-market ticker i we fitted a JumpHMM marginal to the asset's own growth-rate history, using the default configuration ($N = 100$ Laplace-partition states, Student-t emissions with $\nu = 5$, annualized excess growth rates with risk-free rate $r_f = 0$ and $\Delta t = 1/252$); jump parameters (ϵ, λ) were left at the package defaults rather than tuned per ticker, which would improve per-ticker marginal fit but was not required for a comparison that tests the effect of the composition rule on top of whatever marginal the generator produces. For each non-market ticker we then ran OLS against the market path to obtain the calibrated SIM intercept, slope, coefficient of determination, and residual standard deviation. The universe spanned $\beta \in [0.01, 1.79]$ with median $\beta = 1.00$ and quartiles $(0.81, 1.00, 1.18)$, with $R^2_{i,\mathrm{real}}$ ranging from 0.001 to 0.933 (Fig. S3).

The composition rules used a *paired innovation* protocol where applicable: for each ticker and each replication, a single innovation path drawn from the ticker's JumpHMM marginal was passed unchanged to the naive, Gaussian-residual-SIM, and hybrid composers, so that differences among those three reflected the composition rule alone, not generator stochasticity. The *Naive* composer plugged the JumpHMM draw into the SIM identity with no variance correction. The *Gaussian SIM* baseline replaced the JumpHMM draw with i.i.d. Gaussian noise scaled to the calibrated residual variance, recovering R^2 exactly but discarding any marginal structure. The *Hybrid* composer applied Algorithm 1 with floor $f = 0.10$ and threshold $R^2_{\mathrm{preserve}} = 0.80$. Three further composers drew their innovations from dedicated residual generators fit to the real OLS residual series: *JumpHMM-on-residuals* fitted a fresh JumpHMM marginal to the residuals via a pseudo-price inversion; *Block bootstrap* [24] drew Politis–Romano stationary bootstrap replicates of the empirical residual series with mean block length $\sqrt{T} \approx 50$; *GARCH(1,1)-t* [7] fitted a conditional-variance model per ticker, with 30 of the 423 non-stationary fits excluded from that composer only. We skipped the optional cross-sectional rank reorder in Algorithm 1 so per-ticker metrics reflected the marginal construction alone; cross-sectional dependence under copula reorder is outside the scope of the per-ticker evaluation reported here.

For every composed series we computed three metric families against the ticker's real growth-rate history: SIM recovery via OLS on the market path, reporting the absolute deviations $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\mathrm{real}}|$; marginal fidelity via Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) two-sample p-values against the real marginal, plus Wasserstein-1 distance, excess kurtosis, and the Hill upper-tail index on the upper 5% order statistics of $|g|$ (definitions in Online Appendix C); and variance preservation via the ratio of composed to generator sample variance. Pass rates were reported at $\alpha = 0.05$, with $R = 100$ replications per ticker yielding aggregates over $42{,}300$ per-ticker observations per composer; the simulation seed was fixed at 1234.

With the protocol established, we computed per-asset medians of each metric and aggregated across the universe (Table 2). All composers recovered β to within a median absolute error of 0.018 to 0.022, as expected since every composer respects the SIM identity in expectation. The differences between composers appeared everywhere else. The Gaussian SIM baseline recovered R^2 most tightly (median error 0.008) because it targeted R^2 by construction, but it failed marginal fidelity badly: KS pass rate 0.6%, AD pass rate 0.5%, excess kurtosis 1.4 (real data, ~ 13), Hill tail index 0.22 (versus ~ 0.37 on the generator; Fig. S2). The naive composition retained the generator marginal but inflated its variance by $\beta_i^2 \sigma_m^2$, so its KS and AD pass rates sat at 4.5% and 2.7%. The hybrid composer held both properties at once: KS pass rate 50.4%, AD pass rate 47.6%, kurtosis 7.2, Hill 0.36; the R^2 recovery (0.021) came in tighter than the naive baseline but looser than the Gaussian baseline, reflecting that the variance-preserving branch targeted $\sigma^2_{\mathrm{gen},i}$ rather than $R^2_{i,\mathrm{real}}$. The three residual-fit composers sidestepped the variance-inflation problem entirely by fitting a generator directly to e_i, which by construction has variance $\sigma^2_{\varepsilon,\mathrm{real},i}$ and no double-counted market component; their aggregate KS pass

Table 2: **Aggregate scorecard across 423 non-market tickers and 100 replications per ticker.** Median values per composer. $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\text{real}}|$: absolute deviation between OLS recovery and calibration. KS pass and AD pass: fraction of replications whose two-sample test against the real marginal exceeds $p = 0.05$. W_1: Wasserstein-1 distance to the real marginal. κ: sample excess kurtosis. Hill: tail-index estimate on the upper 5% of $|g|$. All three composers recover β to the same precision; only the hybrid composer preserves the marginal distribution (KS pass 50.4%) while holding the variance close to the generator target.

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	AD pass (%)	W_1	κ	Hill
Naive	0.022	0.087	4.5	2.7	0.710	7.767	0.369
Gaussian SIM	0.017	0.008	0.6	0.5	0.682	1.427	0.217
Hybrid	0.018	0.021	50.4	47.6	0.272	7.165	0.365
JumpHMM-on-residuals	0.018	0.015	75.8	70.7	0.232	7.262	0.365
Block bootstrap	0.017	0.020	73.3	66.8	0.232	6.675	0.330
GARCH(1,1)-t	0.017	0.036	67.4	61.4	0.267	6.069	0.317

rates were 75.8% (JumpHMM-on-residuals), 73.3% (block bootstrap), and 66.9% (GARCH(1,1)-t), each higher than hybrid's 50.4%. The advantage sat in the body of the distribution rather than the tails: hybrid's median excess kurtosis (7.17) and Hill upper-tail index (0.365) matched JumpHMM-on-residuals (7.26 and 0.365) to within sampling variability, confirming that the variance correction of Eq. (8) preserves the heavy-tailed character of the full-return generator even when the rescaling is applied; block bootstrap and GARCH, by fitting their own residual generators, reproduced the empirical residual series but with slightly lighter tails ($\kappa = 6.68$ and 6.04; Hill 0.330 and 0.318) than either JumpHMM-based recipe. The downstream consequence of the body-fidelity gap was the question a per-ticker one-day Value-at-Risk backtest was designed to answer (Table 3):

at $\alpha = 0.99$ the nominal exceedance rate is 1%; the naive composer under-breached at 0.67% and the Gaussian baseline over-breached at 1.40%, both failing Kupiec unconditional coverage on the majority of the universe (45 and 47% pass rates), while the hybrid, JumpHMM-on-residuals, block bootstrap, and GARCH(1,1)-t composers all hit coverage within one-tenth of a percentage point of nominal (0.95, 0.99, 1.09, and 1.09% respectively), and their Kupiec pass rates clustered between 72 and 86%. Hybrid and JumpHMM-on-residuals were statistically indistinguishable on this test: the 25 percentage-point body-fidelity gap between them did not translate to a Value-at-Risk accuracy difference. Hybrid's standalone advantage is therefore scoped: it is the right rule when a full-return per-asset generator is the given input, and it delivers tail accuracy on par with a dedicated residual generator without requiring a separate fitting stage.

Having established that body-fidelity did not translate into VaR accuracy, we traced where the KS advantage over the naive baseline came from. We plotted per-ticker KS pass rate and per-ticker marginal variance ratio $\mathrm{Var}(g)/\sigma^2_{\text{gen}}$ as functions of the calibrated β (Fig. 3). The variance-ratio

Table 3: **Per-ticker one-day Value-at-Risk backtest across the six composers.** For each (ticker, replication) pair we estimate the VaR threshold at coverage level α from the composed path and count exceedances against the real 2014–2024 growth-rate history. *rate*: mean exceedance rate across tickers (nominal: $1 - \alpha$). *SD*: cross-ticker standard deviation of the exceedance rate, in percentage points. *Kupiec pass*: fraction of tickers on which the Kupiec unconditional-coverage test p-value exceeds 0.05. Hybrid and JumpHMM-on-residuals are statistically indistinguishable at both coverage levels; both strawman composers (naive, Gaussian SIM) fail coverage on a majority of the universe.

	$\alpha = 0.95$			$\alpha = 0.99$		
Composer	rate (%)	SD (pp)	Kupiec pass (%)	rate (%)	SD (pp)	Kupiec pass (%)
Naive	3.55	0.52	10.4	0.67	0.23	45.2
Gaussian SIM	4.06	0.76	42.6	1.40	0.29	47.0
Hybrid	4.75	0.56	80.4	0.95	0.27	79.7
JumpHMM-on-residuals	4.87	0.55	84.0	0.99	0.26	82.9
Block bootstrap	4.78	0.64	76.7	1.09	0.25	86.0
GARCH(1,1)-t	4.64	0.93	69.0	1.09	0.35	72.1

panel made the mechanism explicit: the naive scatter traced the theoretical curve $1+\rho_i$, overshooting the generator variance by 50% to 75% at high β, while hybrid sat on the unit line by construction. The KS-pass-rate panel showed the consequence: naive fell from a pass rate of ~ 0.25 at $\beta \approx 0.3$ to below 0.01 past $\beta \approx 0.7$, while hybrid sat between 0.3 and 0.9 across the full β range. The Gaussian SIM baseline failed for a different reason: with its Gaussian residual it could not match the heavy-tailed real marginal at any β, collapsing to $\kappa \approx 1$ and Hill ≈ 0.22 regardless of market loading (Fig. S2). Naive and hybrid clustered together near the real-data kurtosis reference line ($\kappa_{\mathrm{real}} \approx 13$) on tail metrics, with the difference between them reflecting the variance inflation under naive rather than any difference in tail shape. Applying the branch-selection rule of Algorithm 1 to the 423-ticker universe assigned 421 tickers to the variance-preserving branch, 2 to the R^2-preserving branch, and none to the clipped variance-preserving branch (Fig. S5, Table S2); the two R^2-preserving tickers were QQQ ($R^2 = 0.861$) and SPYG ($R^2 = 0.933$), both ETFs that track market-capitalization indices closely, and the highest-R^2 individual stock (BLK at $R^2 = 0.645$) sat ~ 0.22 below the $R^2_{\mathrm{preserve}} = 0.80$ threshold, so the branch assignment was empirically tracker-specific on this universe rather than a threshold artifact. The two trackers' KS pass rate under hybrid composition was 99%, consistent with the R^2-preserving branch recovering both the calibrated β and the calibrated R^2 by construction. Clipping did not trigger anywhere in the universe: the market loading ratio ρ_i stayed below $1 - f = 0.90$ for every ticker, because the JumpHMM marginals carried enough variance that the market component never crowded out the $f = 0.10$ idiosyncratic floor. Because the R^2-preserving branch had only two empirical assignments, we validated it on synthetic trackers where the target (β, R^2) was known by construction: across 15 cells of $(\beta_{\mathrm{true}}, R^2_{\mathrm{true}}) \in \{0.8, 1.0, 1.2\} \times \{0.80, 0.85, 0.90, 0.95, 0.99\}$, the branch selector recovered median $\hat{\beta}$ and median $\hat{R}^2$ to within 0.001 of their targets, and cleared the $\alpha = 0.05$ KS test on at least 99% of replications per cell. To check whether the preservation advantage held across the β distribution, we bucketed the universe by β quartile and recomputed the aggregate metrics within each bucket (Table S3): the hybrid KS pass rate declined gently from 67% in the low-β quartile to 44% in the high-β quartile, reflecting the caveat noted alongside Algorithm 1 that as ρ_i approaches the clipping threshold the market-driven component of g tightens the marginal toward Gaussian along its own axis; the effect was small enough that hybrid still beat both baselines by more than an order of magnitude in every bucket. To probe the clipping branch, which did not trigger anywhere in the main run, we re-ran the hybrid composer on the same universe with the market growth-rate path rescaled by $\gamma \in \{1, 2, 3\}$, holding the calibrated marginals fixed. At $\gamma = 1$ no ticker crossed the threshold and the hybrid KS pass rate matched the headline value (50.3%); at $\gamma = 2$ and $\gamma = 3$ the clipping branch activated on 76.4% and 95.2% of tickers respectively, and the hybrid KS pass rate rose to 71.8% and 77.3%. The clipping branch is therefore not a degradation under stress but a mechanism that keeps the composed marginal near the generator target as the market loading inflates.

Finally, we asked whether the headline results were robust to the two hyperparameter choices of Algorithm 1 and to the simulation seed. We re-ran the hybrid composer on the full 423-ticker universe at every point of a 5×5 grid, $R^2_{\mathrm{preserve}} \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$ and $f \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ (Fig. S6); hybrid KS pass rate ranged from 50.24% to 50.38% across the 25 cells, a spread of 0.14 percentage points, with the R^2-preserve assignment stable at QQQ and SPYG for every threshold at or below $R^2 = 0.85$ and the floor f not changing any branch assignment in the grid. The headline number was therefore invariant to the hyperparameter values within wide tolerances, and the operating point ($R^2_{\mathrm{preserve}} = 0.80,\ f = 0.10$) reflects a principled choice of the R^2-branch threshold and a standard floor value rather than a data-peeked local optimum. We also quantified the contribution of generator stochasticity to the headline numbers by re-running the full six-composer protocol under 8 random seeds (Table S4). The per-composer KS pass rates held to within fractions of a percentage point: hybrid sat at $50.40 \pm 0.18\%$ across seeds, JumpHMM-on-residuals at $75.61 \pm 0.11\%$, block bootstrap at $73.24 \pm 0.09\%$, GARCH(1,1)-t at $66.96 \pm 0.30\%$, naive at $4.41 \pm 0.07\%$, and Gaussian SIM at $0.60 \pm 0.04\%$. The across-composer differences therefore exceeded the seed-induced spread by roughly two orders of magnitude, and the seed sweep reproduced the relative ordering of the composers without inversions across seeds, confirming that the hybrid-vs-baseline gap and the residual-fit-vs-hybrid body-fidelity gap were not artifacts of the simulation seed.

5 Discussion

The per-asset evaluation positioned HMM-WJ as the only generator in the eight-model comparison to make meaningful progress on distributional and temporal fidelity at once. Bootstrap resampling set the non-parametric ceiling on marginal fit but inherited the historical realization wholesale and could not extrapolate beyond it. GARCH(1,1) carried the best temporal score (ACF-MAE 0.031) yet failed the KS test on 95% of in-sample paths, indicating that conditional-variance dynamics on their own did not reproduce the empirical body; the same model's Wasserstein-1 distance to the real marginal deteriorated by 67% from in-sample to out-of-sample, exposing a tail misspecification that binary pass rates had obscured. The Gated Recurrent Unit captured a comparable fraction of the temporal structure with 37,954 estimated parameters but produced a 0.6% KS pass rate. The hidden semi-Markov model passed KS at 82%, but its eight-state quantile partition could not span the full growth-rate range, leaving a 38% kurtosis gap. HMM-WJ paid a small distributional cost relative to the no-jump variant (2–8 percentage points lower on the KS and AD pass rates in-sample) in exchange for becoming the only non-GARCH non-neural model to reduce ACF-MAE meaningfully below the i.i.d. floor: approximately one path in four contained a Poisson jump that sustained $\mathrm{ACF}(|G_t|)$ decay across all lags, while the remaining 76% behaved like the no-jump variant. Out-of-sample on the held-out 2025 history HMM-WJ retained a KS pass rate of 94.4% and excess kurtosis 6.1 against the observed 6.9, with the relative ordering of the eight generators preserved.

Three caveats applied to the per-asset stage. First, the state resolution was sized at $N = 100$ Laplace-partition states by a sensitivity sweep over $\{30, 60, 90, 100, 150, 200\}$, beyond which the count of empirical visits per state thinned, and at $N = 350$ certain states received no support in the historical record; for assets with shorter trading histories or with thinner sampling at the extremes, the practical upper bound on state resolution will be lower, and the partition has to be re-sized rather than carried over directly from the SPY calibration. Second, the jump parameters (ϵ, λ) were tuned once against the in-sample SPY history and applied across the 424-ticker universe; the cross-asset extension in Online Appendix G confirmed that the duration parameter λ^* adapted automatically to each asset's clustering intensity at the optimum, but per-ticker tuning of (ϵ, λ) would tighten each marginal at the cost of a calibration step per asset, an exchange we made explicit by reporting universe-wide results under a shared optimum. Third, the Student-t emission with $\nu = 5$ closed the kurtosis gap from 29% under Normal emissions to within 1.3% of the observed value, an improvement that came from matching tail mass at the per-state innovation level. The negative skewness reproduced by both HMM variants (-0.75 observed) came from the asymmetric state-occupancy of the Laplace partition rather than from any explicit skewness parameter, so assets whose growth-rate distribution is skewed in a direction the partition cannot induce will not be matched by this construction without a corresponding adjustment to the quantile rule.

With the per-asset marginals characterized, we turned to the multi-asset stage and asked whether those marginals plugged into a single-index composition rule without losing what made them faithful. We composed synthetic per-asset growth-rate paths under a single-index model (SIM) by drawing each asset's innovation from an independent JumpHMM generator fit to the asset's own full growth-rate history, then rescaling the draw by a closed-form scalar before inserting it into the SIM composition. We evaluated six composers on a calibrated universe of 424 US equities and exchange-traded funds against the real SPY market path: the naive composition, a Gaussian-residual SIM baseline, the hybrid construction proposed here, and three residual-fit alternatives (JumpHMM-on-residuals, Politis–Romano block bootstrap [24], and a per-ticker GARCH(1,1)-t fit [7]). The one-line variance correction, together with the sign-preserving clip rule and the R^2-preserving branch, closes a gap between two previously incomplete options for building synthetic asset paths with both marginal structure and factor structure: naive composition gives both but distorts the marginal by $\beta^2\sigma_m^2$, while Gaussian-residual SIM gives the factor exactly but strips the marginal. The hybrid construction sat between the Gaussian-residual and naive baselines: it recovered β to the same precision as either, held the generator marginal to its target variance in every ticker, and preserved the generator's tails at a level indistinguishable from the naive composition (Fig. 3, Table 2). The three residual-fit composers produced higher Kolmogorov–Smirnov pass rates than hybrid on marginal body fidelity (67 to 76% versus 50%, Table 2), but matched hybrid on the 99% Value-at-Risk coverage test to within statistical tolerance (Table 3); the body-fidelity gap did not translate to a tail-accuracy gap. The contribution of hybrid was therefore scoped to the case where the per-asset generator was specified in full-return units (a common setting, since per-asset generators are often calibrated for distributional interpretability, cached for other pipelines, or chosen to retain regime-switching or jump structure); in that case

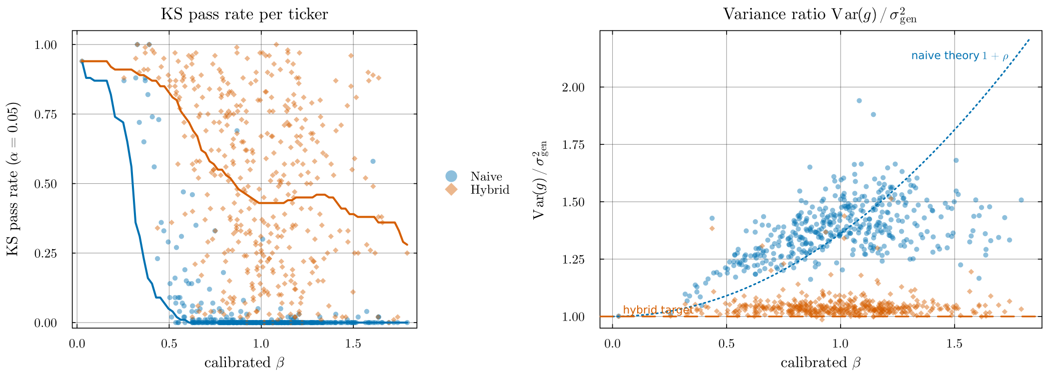

Figure 3: **Hybrid's variance correction and its KS-pass-rate consequence, per ticker vs calibrated β.** Each point is a per-ticker median over 100 replications; thick lines are binned medians. *Left (consequence):* Kolmogorov–Smirnov pass rate against the real marginal at $\alpha = 0.05$. The naive composer (blue) collapses past $\beta \approx 0.6$ as market loading inflates the composed marginal; the hybrid composer (orange) stays between ~ 0.3 and ~ 0.9 across the full β range. *Right (mechanism):* variance ratio $\mathrm{Var}(g)/\sigma^2_{\mathrm{gen}}$. The dotted curve is the theoretical naive inflation $1 + \rho$ evaluated at the universe-median σ^2_{gen}; the naive scatter follows it, overshooting generator variance by 50 to 75% at high β. Hybrid sits on the unit line (dashed) by construction.

hybrid delivered 99% VaR accuracy on par with a dedicated residual generator without requiring a separate fitting stage. The variance-inflation gap was large enough to matter in practice: at the high-β end of the universe, naive composition inflated the marginal variance by over 70% relative to the generator target (Fig. 3). Because branch selection was driven entirely by the calibrated R^2 and each asset carried a persisted construction flag, the procedure ran uniformly across the ticker universe without external metadata and left an audit trail that downstream consumers could inspect. No part of the construction is new; the contribution is a minimal, practitioner-ready recipe with an auditable branch flag and an explicit treatment of the tracker limit, usable as a drop-in replacement for naive composition in workflows where both tail shape and market dependence matter.

The evaluation isolated the composition rule by fixing the market path at the real SPY history and using the same per-ticker innovation draw across all three composers. This design yielded a paired comparison of composition rules at the per-ticker level but set aside two questions that production users will need to address separately. The first concerns cross-sectional dependence: we skipped the optional rank-reorder step in Algorithm 1 so that per-ticker metrics reflected the composition rule alone. When the reorder is applied, the variance algebra is unchanged (rank-reordering preserves the marginal variance of ε_i), so the marginal results here carry over; cross-sectional fidelity under different copula specifications is a separate evaluation that we defer to follow-up work. The second concerns market-path uncertainty: the metrics reported here used the real SPY path, and under a simulated market path the composed marginal picks up additional variance from the sampling of $g_m^{(r)}$, which for high-R^2 tickers becomes a noticeable effect. In production, the two deferred questions compose: a reorder-plus-simulated-market pipeline would layer cross-sectional structure on top of per-ticker draws built against sampled market paths, with the variance correction of Eq. (8) still providing the per-asset variance target on each realized $g_m^{(r)}$. A separate scope boundary concerns the downstream use case: the main metrics reported here characterize marginal and factor fidelity of the composition rule on per-ticker targets. To test whether that fidelity translates to a consequential downstream number, we also report a per-ticker one-day 95% and 99% Value-at-Risk backtest of the composed paths against the real 2014–2024 growth-rate histories, with exceedance counts assessed by Kupiec's unconditional coverage test; the per-ticker framing keeps this validation inside the paper's scope, while portfolio-level VaR requires the cross-sectional copula reorder deferred above. High-turnover portfolio strategies remain a distinct scope boundary: the temporal structure of the residual draw matters in a way that contemporaneous composition does not capture.

A related boundary on the recipe was the clipping rule. Clipping did not trigger anywhere in the US-equity universe used here, because the JumpHMM marginals carried enough variance that $\rho_i = \beta_i^2 \sigma_m^2 / \sigma^2_{\mathrm{gen},i}$ stayed below 0.90 even for the highest-β ticker. Clipping becomes relevant in two settings we did not evaluate: scenario-driven markets, where the synthetic g_m is constructed to be much more volatile than the calibration market (e.g., stress tests), and universes that mix low-volatility single stocks with extreme betas (e.g., leveraged ETFs). In both settings we recommend reporting the original and effective β_i alongside the output, as in Algorithm 1, so that downstream consumers can audit the attenuation. A related attenuation appeared at the other end of the β spectrum, even when clipping did not trigger: the hybrid KS pass rate fell from 67% at low β to 44% at high β (Table S3), a mild attenuation that followed directly from the composition algebra. As $\beta_i^2\sigma_m^2$ grows toward the idiosyncratic floor $(1-f)\sigma^2_{\mathrm{gen},i}$, the market-driven component carries an increasing share of the marginal, and that component is only as heavy-tailed as g_m. The market's own heavy tails slow the attenuation (SPY growth rates are themselves heavy-tailed after JumpHMM fitting), but the effect is visible. Users who need exact marginal fidelity at high β can recover it by lowering the floor f (at the cost of less conservative clipping behavior) or by pre-scaling the generator variance so that ρ_i stays small.

The variance correction $s_i^2 = 1 - \rho_i$ relies on $\mathrm{Cov}(\tilde{g}_i, g_m) \approx 0$. The JumpHMM marginals are fit in isolation and sampled independently of the market path, so this holds by construction; if the generator is already correlated with the market (for example, a bootstrap from joint history), the empirical covariance should be subtracted from β_i before applying the correction, or s_i^2 should be estimated directly from the empirical variance of the composed series. The same warning applies to the R^2-preserving branch, whose variance identity assumes ε is uncorrelated with g_m after any reorder. The construction also injects a contemporaneous market factor only: $g_i(t)$ depends on $g_m(t)$ and on $\tilde{g}_i(t)$ but not on $g_m(t-1), g_m(t-2), \ldots$, so lagged market dependence (lead-lag effects, market-driven volatility clustering, asymmetric responses to drawdowns) is not reproduced by this recipe. The natural extension is a lagged-factor SIM, $g_i(t) = \alpha_i + \sum_{k=0}^{K} \beta_{i,k}\, g_m(t-k) + \varepsilon_i(t)$; the variance correction generalizes to that form by replacing $\beta_i^2\sigma_m^2$ with the variance of the full market-factor component, though the empirical value of the extra parameters is beyond the scope here. Lagged idiosyncratic dependence is likewise absent: each generator growth-rate path $\tilde{g}_i$ is sampled independently of the others, so any residual autocorrelation present in the real data is discarded by the composition. This matters most for workflows that evaluate portfolios rebalancing at high frequency. With iid residuals, a rebalancing rule collects the full diversification return of [44], a compounding premium that scales with residual variance and rebalancing frequency and is largely absent in real markets, where daily residual autocorrelation is small but positive [50]. Two generator alternatives in Table 2 preserve temporal structure directly: the block-bootstrap composer resamples blocks of the empirical OLS residual series [24], inheriting whatever autocorrelation is present in the data, and the GARCH(1,1)-t composer fits a conditional-variance model per ticker [7]. Both recover marginal fidelity at the cost of committing the pipeline to a specific residual generator, whereas the variance correction of Eq. (8) is generator-agnostic and accepts a full-return JumpHMM marginal (with its jump and regime structure) as input. Users evaluating high-turnover strategies on JumpHMM-corrected paths should benchmark absolute return levels against a static-allocation baseline on the same paths, or adopt the bootstrap or GARCH alternatives when residual autocorrelation is the relevant structure for the downstream task.

The diversification-return artifact is quantitatively large enough on standard rebalancing rules that absolute-return claims drawn from JumpHMM-corrected paths require an explicit correction or an explicit caveat. As a concrete demonstration, a 20-asset target-weight allocator deployed against a hybrid-composed 2025 universe (calibration window 2014–2024) produced a median synthetic continuously-compounded growth rate (CCGR) of $\sim 30\%/\mathrm{yr}$, while the same allocator deployed against the real 2025 growth-rate history produced $11.5\%/\mathrm{yr}$. The decomposition isolated the artifact: equal-weight buy-and-hold on the synthetic ensemble produced $\sim 7.7\%/\mathrm{yr}$, the same target-weight allocation held statically (no rebalance) produced $\sim 2.0\%/\mathrm{yr}$, and switching the same allocation rule to daily rebalancing produced $\sim 29.7\%/\mathrm{yr}$, a $+27.7$ percentage-point jump from the static-to-daily switch on the same allocation rule and the same paths (Online Appendix S2). The $\sim 22\,\mathrm{pp}$ gap between the synthetic and real engine CCGR was therefore a structural feature of the iid-residual generator family rather than a calibration error, consistent with [44, 50, 51].

Two practical responses follow. Where the workflow can tolerate being explicit about the issue, we recommend reporting the static-versus-rebalanced decomposition on the same allocation rule and the same synthetic paths, and framing absolute Sharpe levels under daily rebalancing as "this generator with this strategy" rather than as a forward prediction. Where the workflow requires an absolute-return level for downstream consumption (e.g., percentile diagnostics overlaying real outcomes against a synthetic distribution), we recommend a uniform location-shift bias correction (Eq. S7, Online Appendix S2) that scales each synthetic wealth path by an exponential drag whose rate is set so the corrected median annualized growth rate matches a documented long-run prior, typically near eight percent per year, decomposable into the prevailing risk-free rate plus an active premium net of frictions [52, 53]. The correction is a per-path monotone transform: it preserves the rank order of paths by terminal wealth, the drawdown shape, and the volatility envelope; only the level shifts. Anchoring to a documented prior rather than to the realized test-year outcome keeps the correction independent of the sample to which the synthetic ensemble is being compared and avoids circular calibration. The realized year then lands inside the corrected ensemble at an honest percentile (in our deployment, real-2025 landed at the 63rd percentile of the corrected ensemble, consistent with 2025 being an above-typical equity year). Online Appendix S2 details the methodology, contrasts the location-shift correction with a multiplicative haircut alternative (which we do not recommend because it scrambles per-path rank order and distorts the drawdown distribution), and discusses why several intuitive alternatives (parameter drift between calibration and deployment, transaction-cost inflation, AR(1) on residuals without restructuring the copula sampling, residual block-bootstrap) do not address the underlying mechanism within the iid-residual SIM framework. The synthetic data remained reliable, both before and after correction, for the ensemble-shape questions the paper validates: percentile structure, drawdown distribution, and relative ranking of strategies on the same paths. The location-shift correction extends that reliability to absolute Sharpe and CCGR levels under daily rebalancing, at the cost of an external prior.

Two further extensions follow naturally. Multi-factor SIM replaces the single market factor with K factors, requiring ρ_i to become the sum of factor contributions and the loading vector to be clipped rather than a scalar, though the branch logic is unchanged. A stress-test harness pairs the clipping rule with a constructed market scenario more volatile than history: enforcing a minimum idiosyncratic share guarantees that per-asset marginals remain heavy-tailed in precisely the regime where naive SIM composition breaks down most severely.

6 Conclusion

We presented a two-stage construction for synthetic equity path generation that closes two gaps in the literature: a hybrid hidden Markov framework that reproduces single-asset stylized facts via a Laplace quantile partition and a Poisson jump-duration override, and a closed-form variance-corrected single-index composer that places those validated marginals in a multi-asset factor structure without inflating the marginal variance. The construction is generator-agnostic on the composition side, in the sense that any per-asset full-return generator can plug into the variance correction and inherit the marginal it was built to deliver.

On the empirical side, the hybrid HMM marginals passed the Kolmogorov-Smirnov test against the real SPY univariate distribution at 97.6% in-sample and 94.4% out-of-sample, outperforming Bootstrap, Gaussian, Laplace, GARCH, hidden semi-Markov, and Gated Recurrent Unit baselines on combined distributional and tail metrics. The variance-corrected single-index composer recovered the factor loading to a median absolute error of 0.018 and matched the leading residual-fit alternatives on a per-ticker one-day 99% Value-at-Risk backtest, validating the composed paths against the real 2014–2024 growth-rate histories. We documented one downstream caveat: an iid-residual diversification-return artifact under daily rebalancing that inflates absolute return levels by roughly 22 percentage points on a deployed 20-asset target-weight allocator. A uniform location-shift bias correction anchored to a documented long-run prior addresses the artifact as a per-path monotone transform that preserves rank order, drawdown shape, and volatility envelope while shifting only the level.

Several extensions follow naturally from the present work. The variance correction generalizes to multi-factor and lagged-factor single-index models without changing its essential algebra; we leave the empirical evaluation of these extensions to follow-up work. Per-ticker tuning of the jump parameters (ϵ, λ) would push the hybrid composer's marginal-fidelity pass rates upward at the cost of a calibration step per asset. The diversification-return artifact admits a structural fix via autoregressive or block-bootstrap residuals composed with the single-index factor; that fix requires restructuring the copula sampling step and is the most consequential follow-up for high-turnover downstream workflows.

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Author contributions. A.A. designed the hybrid hidden Markov marginal generator, performed the single-asset experiments and validation framework, and contributed to the manuscript. J.D.V. designed the variance-corrected single-index composition, performed the multi-asset experiments and downstream Value-at-Risk validation, developed the bias-correction methodology, released the code, and contributed to the manuscript.

Competing interests. The authors declare no competing interests.

Code and data availability. All source code, experiment scripts, and fitted per-asset marginals are released as part of `JumpHMM.jl` (https://github.com/varnerlab/JumpHMM.jl); the paper-specific evaluation harness (six-composer benchmark, Value-at-Risk backtest, bias-correction utility) lives in the `paper/` subdirectory of that repository.

References

[1] Samuel A. Assefa, Danial Dervovic, Mahmoud Mahfouz, Robert E. Tillman, Prashant Reddy, and Manuela Veloso. Generating synthetic data in finance: Opportunities, challenges and pitfalls. In *Proceedings of the First ACM International Conference on AI in Finance*, pages 1–8, 2020. doi: 10.1145/3383455.3422554.

[2] James Jordon, Lukasz Szpruch, Florimond Houssiau, Mirko Bottarelli, Giovanni Cherubin, Carsten Maple, Samuel N. Cohen, and Adrian Weller. Synthetic data: What, why and how? *arXiv preprint arXiv:2205.03257*, 2022.

[3] Benoit Mandelbrot. The variation of certain speculative prices. *The Journal of Business*, 36(4): 394–419, 1963.

[4] R. Cont. Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236, 2001. doi: 10.1080/713665670.

[5] Michael Stenger, André Bauer, Thomas Prantl, Robert Leppich, Nathaniel Hudson, Kyle Chard, Ian Foster, and Samuel Kounev. Thinking in categories: A survey on assessing the quality for time series synthesis. *Journal of Data and Information Quality*, 16(2):1–32, 2024. doi: 10.1145/3666006.

[6] Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

[7] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986. doi: 10.1016/0304-4076(86)90063-1.

[8] Steven L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2):327–343, 1993. doi: 10.1093/rfs/6.2.327.

[9] Robert C. Merton. Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1-2):125–144, 1976. doi: 10.1016/0304-405X(76)90022-2.

[10] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9 (8):1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.

[11] Thomas Fischer and Christopher Krauss. Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654–669, 2018. doi: 10.1016/j.ejor.2017.11.054.

[12] Shuntaro Takahashi, Yu Chen, and Kumiko Tanaka-Ishii. Modeling financial time-series with generative adversarial networks. *Physica A: Statistical Mechanics and Its Applications*, 527: 121261, 2019. doi: 10.1016/j.physa.2019.121261.

[13] Sohyeon Kwon and Yongjae Lee. Can GANs learn the stylized facts of financial time series? In *Proceedings of the 5th ACM International Conference on AI in Finance*, pages 126–133, 2024. doi: 10.1145/3677052.3698661.

[14] Jinsung Yoon, Daniel Jarrett, and Mihaela van der Schaar. Time-series generative adversarial networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[15] L. Rabiner and B. Juang. An introduction to hidden Markov models. *IEEE ASSP Magazine*, 3 (1):4–16, 1986. doi: 10.1109/MASSP.1986.1165342.

[16] Tobias Rydén, Timo Teräsvirta, and Stefan Åsbrink. Stylized facts of daily return series and the hidden Markov model. *Journal of Applied Econometrics*, 13(3):217–244, 1998. doi: 10.1002/(SICI)1099-1255(199805/06)13:3<217::AID-JAE476>3.0.CO;2-V.

[17] Jan Bulla and Ingo Bulla. Stylized facts of financial time series and hidden semi-Markov models. *Computational Statistics & Data Analysis*, 51(4):2192–2209, 2006. doi: 10.1016/j.csda.2006. 07.021.

[18] William F. Sharpe. A simplified model for portfolio analysis. *Management Science*, 9(2): 277–293, 1963. doi: 10.1287/mnsc.9.2.277.

[19] Stephen A. Ross. The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13 (3):341–360, 1976. doi: 10.1016/0022-0531(76)90046-6.

[20] Eugene F. Fama and Kenneth R. French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, 1993. doi: 10.1016/0304-405X(93)90023-5.

[21] Leonard E. Baum, Ted Petrie, George Soules, and Norman Weiss. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics*, 41(1):164–171, 1970. doi: 10.1214/aoms/1177697196.

[22] Jeff A. Bilmes. A gentle tutorial of the EM algorithm and its application to parameter estimation for Gaussian mixture and hidden Markov models. Technical Report TR-97-021, International Computer Science Institute, 1998.

[23] Kyunghyun Cho, Bart van Merrienboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder-decoder for statistical machine translation. *arXiv preprint arXiv:1406.1078*, 2014.

[24] Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994. doi: 10.1080/01621459.1994.10476870.

[25] Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.

[26] Eugene F. Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417, 1970. doi: 10.2307/2325486.

[27] G. William Schwert. Why does stock market volatility change over time? *The Journal of Finance*, 44(5):1115–1153, 1989. doi: 10.1111/j.1540-6261.1989.tb02647.x.

[28] Francis X. Diebold and Atsushi Inoue. Long memory and regime switching. *Journal of Econometrics*, 105(1–2):131–159, 2001. doi: 10.1016/S0304-4076(01)00073-2.

[29] Elias M. Stein and Jeremy C. Stein. Stock price distributions with stochastic volatility: An analytic approach. *The Review of Financial Studies*, 4(4):727–752, 1991.

[30] S. G. Kou. A jump-diffusion model for option pricing. *Management Science*, 48(8):1086–1101, 2002. doi: 10.1287/mnsc.48.8.1086.166.

[31] Jay F. K. Au Yeung, Zi-kai Wei, Kit Yan Chan, Henry Y. K. Lau, and Ka-Fai Cedric Yiu. Jump detection in financial time series using machine learning algorithms. *Soft Computing*, 24(3): 1789–1801, 2020. doi: 10.1007/s00500-019-04006-2.

[32] James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989. doi: 10.2307/1912559.

[33] Alessandro Rossi and Giampiero M. Gallo. Volatility estimation via hidden Markov models. *Journal of Empirical Finance*, 13(2):203–230, 2006. doi: 10.1016/j.jempfin.2005.09.003.

[34] Nguyet Nguyen. Hidden Markov model for stock trading. *International Journal of Financial Studies*, 6(2):36, 2018. doi: 10.3390/ijfs6020036.

[35] Chang-Jin Kim and Charles R. Nelson. Has the U.S. economy become more stable? A Bayesian approach based on a Markov-switching model of the business cycle. *The Review of Economics and Statistics*, 81(4):608–616, 1999. doi: 10.1162/003465399558472.

[36] David Berghaus, Kostadin Cvejoski, Patrick Seifner, César Ojeda, and Ramsés J. Sanchez. Foundation inference models for Markov jump processes. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*, pages 129407–129442, 2024.

[37] José G. Dias, Jeroen K. Vermunt, and Sofia Ramos. Mixture hidden Markov models in finance research. In *Advances in Data Analysis, Data Handling and Business Intelligence*, pages 451–459. Springer, 2009. doi: 10.1007/978-3-642-01044-6_41.

[38] Paul Embrechts, Alexander J. McNeil, and Daniel Straumann. Correlation and dependence in risk management: Properties and pitfalls. In *Risk Management: Value at Risk and Beyond*, pages 176–223. Cambridge University Press, 2002. doi: 10.1017/CBO9780511615337.008.

[39] Umberto Cherubini, Elisa Luciano, and Walter Vecchiato. *Copula Methods in Finance*. John Wiley & Sons, 2004. ISBN 9780470863459.

[40] Kjersti Aas, Claudia Czado, Arnoldo Frigessi, and Henrik Bakken. Pair-copula constructions of multiple dependence. *Insurance: Mathematics and Economics*, 44(2):182–198, 2009.

[41] A. N. Kolmogorov. Sulla determinazione empirica di una legge di distribuzione. *Giornale dell'Istituto Italiano degli Attuari*, 4:83–91, 1933.

[42] N. Smirnov. Table for estimating the goodness of fit of empirical distributions. *The Annals of Mathematical Statistics*, 19(2):279–281, 1948.

[43] Theodore W. Anderson and Donald A. Darling. Asymptotic theory of certain "goodness of fit" criteria based on stochastic processes. *The Annals of Mathematical Statistics*, 23(2):193–212, 1952.

[44] David G. Booth and Eugene F. Fama. Diversification returns and asset contributions. *Financial Analysts Journal*, 48(3):26–32, 1992. doi: 10.2469/faj.v48.n3.26.

[45] Andrew W. Lo and A. Craig MacKinlay. When are contrarian profits due to stock market overreaction? *Review of Financial Studies*, 3(2):175–205, 1990. doi: 10.1093/rfs/3.2.175.

[46] Samuel Kotz, Tomasz J. Kozubowski, and Krzysztof Podgórski. *The Laplace Distribution and Generalizations: A Revisit with Applications to Communications, Economics, Engineering, and Finance*. Birkhauser, 2001. ISBN 9780817641665.

[47] Paul Glasserman. *Monte Carlo Methods in Financial Engineering*. Springer, 2003. doi: 10.1007/978-0-387-21617-1.

[48] James D. Hamilton. Regime switching models. *The New Palgrave Dictionary of Economics*, pages 11421–11426, 2018. doi: 10.1057/978-1-349-95189-5_2459.

[49] Charles M. Grinstead and J. Laurie Snell. *Introduction to Probability*. American Mathematical Society, 1997.

[50] Paul Bouchey, Duncan Gunthorp, and Matt Sutherland. Volatility harvesting in theory and practice. *Journal of Wealth Management*, 15(2):89–100, 2012.

[51] Harry M. Markowitz. Investment for the long run: New evidence for an old rule. *Journal of Finance*, 31(5):1273–1286, 1976. doi: 10.1111/j.1540-6261.1976.tb03213.x.

[52] Aswath Damodaran. Equity risk premiums: Determinants, estimation and implications. NYU Stern School of Business working paper, updated annually, 2025. https://pages.stern.nyu.edu/~adamodar/.

[53] Roger G. Ibbotson and Duff and Phelps. *SBBI Yearbook: Stocks, Bonds, Bills, and Inflation*. Duff and Phelps / Kroll, annual edition, 2024.

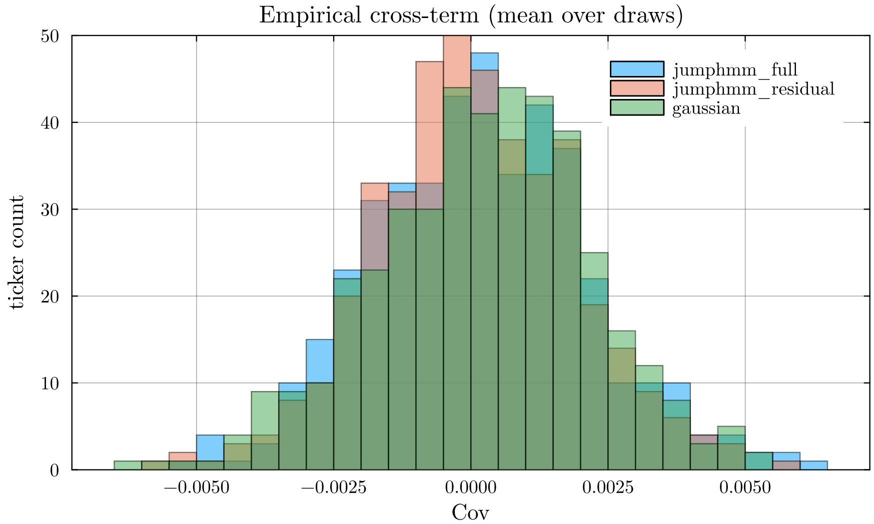

Figure S1: **Empirical normalized cross-covariance** $\mathrm{Cov}[\tilde{g}_i, g_m]/(\sigma_{\mathrm{gen},i}\,\sigma_m)$ **across the** 423**-ticker universe.** Histograms were stacked per generator, with each per-ticker value averaged over $R = 100$ generator growth-rate paths. The full-return JumpHMM, residual-fit JumpHMM, and Gaussian generators all clustered at zero, validating the independence assumption used to derive Eq. (8).

A Empirical cross-term diagnostic

The variance correction of Eq. (8) rests on the assumption $\mathrm{Cov}[\tilde{g}_i, g_m] \approx 0$, which is what allows the derivation to drop the cross-term in the expansion of $\mathrm{Var}[g_i]$. Because the per-asset JumpHMM marginals are fit to each asset's full growth-rate history, including its market-correlated component, the assumption is not automatic: a generator that picked up a residual market loading would invalidate the closed-form scaling. We tested the assumption empirically by sampling $R = 100$ generator growth-rate paths $\tilde{g}_i^{(r)}$ per ticker from each generator and computing the normalized cross-covariance against the real SPY path:

$$\mathrm{cov}_i^{\mathrm{norm}} = \frac{1}{R}\sum_{r=1}^{R} \frac{\mathrm{Cov}[\tilde{g}_i^{(r)},\, g_m]}{\sigma_{\mathrm{gen},i}\,\sigma_m}. \tag{S1}$$

Per-ticker values were plotted as histograms in Fig. S1. Across the 423 tickers, the full-return JumpHMM generator (used by the hybrid and naive composers) sat within ± 0.006 of zero with a per-ticker mean bias of 1.5×10^{-4} (Fig. S1); the residual-fit JumpHMM generator (used by the JumpHMM-on-residuals composer) and the Gaussian baseline were also centered at zero (mean 7×10^{-5} and 2.3×10^{-4} respectively, Fig. S1). The independence assumption was therefore satisfied to within sampling tolerance for all three generators, and the derivation of Eq. (8) held in practice.

B Bias correction for daily-rebalanced backtests

The synthetic paths described in the main text reproduce per-asset marginals and single-factor cross-sectional dependence faithfully but discard temporal structure in the residual draw: at each trading day the per-asset residual $\varepsilon_i(t)$ is sampled independently from the previous day. Real equity residuals carry a small but positive serial autocorrelation [45], typically $\rho \in [0.05, 0.10]$ at the daily horizon, so the iid-residual generator family used here drops a real feature of the data. To make the consequences precise for daily-rebalanced backtests, consider a portfolio of N assets with constant target weights $w_i \geq 0$, $\sum_i w_i = 1$, reset to those weights at the start of each trading day. Throughout this derivation $g_i(t)$ is the annualized log-growth rate of Eq. (1), so the per-period log return on asset i is $g_i(t)\,\Delta t$ with $\Delta t = 1/252$ yr. The wealth process is $W_t = W_{t-1}\sum_i w_i \exp(g_i(t)\,\Delta t)$, and the per-step log return is $\log(W_t/W_{t-1}) = \log \sum_i w_i \exp(g_i(t)\,\Delta t)$. Second-order Taylor expansion of $\exp(g_i\,\Delta t)$

about zero, followed by second-order expansion of $\log(1+x)$ about zero, gives

$$\log\frac{W_t}{W_{t-1}} \;=\; \Delta t\sum_i w_i\, g_i(t) \;+\; \tfrac{1}{2}(\Delta t)^2\left(\sum_i w_i\, g_i(t)^2 \;-\; \Big(\sum_i w_i\, g_i(t)\Big)^2\right) \;+\; O\big((g\,\Delta t)^3\big), \tag{S2}$$

which, after dividing by Δt to extract the annualized single-step rate $r_p(t) := \Delta t^{-1}\log(W_t/W_{t-1})$ and taking unconditional expectations under stationarity (using $\mathbb{E}[g_i^2] = \mu_i^2 + \sigma_i^2$ and $\mathbb{E}[(\sum_i w_i g_i)^2] = (\sum_i w_i \mu_i)^2 + \sigma_p^2$ with $\sigma_p^2 = w^\top \Sigma w$ the per-rate portfolio variance), becomes

$$\mathbb{E}[r_p] \;=\; \sum_i w_i\,\mu_i + \underbrace{\tfrac{1}{2}\,\Delta t\left(\sum_i w_i\,\sigma_i^2 - \sigma_p^2\right)}_{D} + \tfrac{1}{2}\,\Delta t\left(\sum_i w_i\,\mu_i^2 - \Big(\sum_i w_i\,\mu_i\Big)^2\right) + O\big((\Delta t)^2\big), \tag{S3}$$

the Booth-Fama identity for the daily-rebalanced portfolio in annualized growth-rate units [44, 51]. The per-rate moments $\mu_i = \mathbb{E}[g_i(t)]$ and $\sigma_i^2 = \mathrm{Var}(g_i(t))$ carry units yr^{-1} and yr^{-2} respectively; the products $\Delta t\,\sigma_i^2$ and $\Delta t\,\sigma_p^2$ in Eq. (S3) are the annualized variances (yr^{-1}) that finance practice reports directly. The third bracket is the cross-sectional dispersion of the per-asset annualized means, of order $\Delta t\,\mu^2 \sim 3\times10^{-5}\ \mathrm{yr}^{-1}$ at $\mu \sim 0.08\ \mathrm{yr}^{-1}$ and negligible against the variance term $D \sim \Delta t\,\sigma^2 \sim 4\times10^{-2}\ \mathrm{yr}^{-1}$; we drop it. The realized annualized compound growth rate $\bar{g}_p = (T\,\Delta t)^{-1}\log(W_T/W_0) = T^{-1}\sum_t r_p(t)$ converges to $\mathbb{E}[r_p]$ by the ergodic theorem under any stationary residual process, so the long-run limit is fixed by the one-time-slice unconditional joint distribution of $(g_i(t))_i$ and does not, in itself, depend on the serial structure of the residual draw.

The middle term D in Eq. (S3) is the diversification return. Expanding $\sigma_p^2 = \sum_i w_i^2\sigma_i^2 + 2\sum_{i<j} w_i w_j \Sigma_{ij}$ and using the identity $\sum_i w_i(1-w_i)\sigma_i^2 = \sum_{i<j} w_i w_j(\sigma_i^2+\sigma_j^2)$ that holds for any weights summing to one,

$$D \;=\; \tfrac{1}{2}\,\Delta t\left(\sum_i w_i\sigma_i^2 - \sigma_p^2\right) \;=\; \tfrac{1}{2}\,\Delta t\sum_{i<j} w_i w_j\ \mathrm{Var}(g_i - g_j) \;\geq\; 0, \tag{S4}$$

so D equals one-half the annualized weighted-average pairwise return-difference variance and is therefore non-negative, with equality only when every pair (i,j) comoves perfectly ($\mathrm{Var}(g_i - g_j) = 0$) or all weight concentrates on a single asset. The mechanism is the rebalancing rule itself: weights drift between rebalances, the rebalancer sells whatever has outperformed (its weight rose above target) and buys whatever has underperformed (its weight fell below target), and the trade systematically converts period-by-period cross-sectional dispersion into compounded growth. Markowitz [51] described this as the variance bonus of the long-run mean-variance allocation; Booth and Fama [44] called it the diversification return.

Substituting the single-index decomposition $g_i(t) = \alpha_i + \beta_i\, g_m(t) + \varepsilon_i(t)$ with $\mathrm{Cov}[g_m, \varepsilon_i] = 0$ (verified in Sec. A) into Eq. (S4) gives

$$\mathrm{Var}(g_i - g_j) \;=\; (\beta_i - \beta_j)^2\sigma_m^2 \;+\; \mathrm{Var}(\varepsilon_i - \varepsilon_j), \tag{S5}$$

since g_m and the residuals are uncorrelated, so the diversification return splits cleanly into a market-beta-dispersion component and an idiosyncratic component,

$$D \;=\; \underbrace{\tfrac{1}{2}\,\Delta t\,\sigma_m^2\sum_{i<j} w_i w_j(\beta_i - \beta_j)^2}_{D_{\mathrm{mkt}}} \;+\; \underbrace{\tfrac{1}{2}\,\Delta t\sum_{i<j} w_i w_j\ \mathrm{Var}(\varepsilon_i - \varepsilon_j)}_{D_\varepsilon}, \tag{S6}$$

where the first term equals $\frac{1}{2}\,\Delta t\,\sigma_m^2[\sum_i w_i\beta_i^2 - (\sum_i w_i\beta_i)^2]$, the weighted cross-sectional β dispersion scaled by the annualized market variance $\Delta t\,\sigma_m^2$. Across the 423-ticker universe the empirical β distribution concentrates in $[0.5, 1.5]$ (Sec. G) and the SPY annualized variance $\Delta t\,\sigma_m^2 \approx 0.04\ \mathrm{yr}^{-1}$, so D_{mkt} contributes at most a few percentage points per year for any diversified equity portfolio. The idiosyncratic component D_ε is the dominant term: with residuals approximately orthogonal across assets at the same time slice, $\mathrm{Var}(\varepsilon_i - \varepsilon_j) \approx \sigma_{\varepsilon,i}^2 + \sigma_{\varepsilon,j}^2$, and D_ε scales with the weighted-average annualized idiosyncratic variance $\Delta t\,\sigma_{\varepsilon,i}^2$ per name. Concentrated allocators on high-idiosyncratic-volatility names therefore inherit large formula values of D_ε structurally.

Table S1: **Descriptive statistics and stylized-facts tests for SPY daily excess growth rates.** In-sample: 2014–2024 ($T = 2{,}766$); out-of-sample: 2025 ($T = 249$). Gaussian and Laplace columns show maximum likelihood estimation (MLE) fits to the in-sample data. JB: Jarque-Bera normality test. LB: Ljung-Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed	Gaussian	Laplace
Mean (annualized, %)	6.31	11.60	6.31	14.19
Std Dev (annualized, %)	214.50	220.62	214.46	205.93
Skewness	−0.753	−1.093	0	0
Excess Kurtosis	7.715	6.867	0	3
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a
LB test on G_t (lag 20)	reject* ($p < 0.001$)	reject* ($p = 0.019$)	n/a	n/a
LB test on $\lvert G_t \rvert$ (lag 20)	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a

The iid construction of the residual draw fixes both terms of Eq. (S6). The variance correction of Eq. (8) pins the synthetic per-asset variance $\sigma^2_{\mathrm{gen},i} = \beta_i^2 \sigma_m^2 + \sigma^2_{\varepsilon,i}$ to its empirical target, so the formula value $D = D_{\mathrm{mkt}} + D_\varepsilon$ on synthetic paths equals the formula value under any alternative residual generator matched on the same unconditional second moments, including a stationary AR(1) residual generator $\varepsilon_i(t) = \phi\, \varepsilon_i(t-1) + \eta_i(t)$ with iid innovations η_i tuned so that $\mathrm{Var}(\varepsilon_i) = \sigma^2_{\eta,i}/(1-\phi^2) = \sigma^2_{\varepsilon,i}$. The AR(1) parameter ϕ leaves the unconditional joint distribution of $(g_i(t))_i$ at a single time slice unchanged, which by Eqs. (S3), (S6) means ϕ enters at zeroth order in $\mathbb{E}[r_p]$ and at first order only in the long-run variance of the sample mean ($\mathrm{Var}(\bar{g}_p) \to T^{-1}\, \sigma_p^2\, (1+\phi)/(1-\phi)$). The synthetic-ensemble median across 1,000 paths therefore converges to the same formula value whether the residuals are drawn iid or with positive serial autocorrelation: the iid versus AR(1) distinction shows up in per-path sampling noise, not in the ensemble median itself. The empirical excess of the engine-raw median over an external long-run growth prior, reported quantitatively as $\sim$22 percentage points in the next paragraph, is therefore not a structural artifact of the iid sampling at the level of the second-order formula. It is the combined empirical magnitude of the higher-moment corrections to the Taylor expansion (heavy-tailed JumpHMM marginals carry non-trivial third and fourth cumulants), the time-varying volatility and correlation across the 2014–2024 calibration window relative to a long-run prior, and the documented attenuation of realized diversification returns on real equity portfolios at all rebalancing horizons [50]. The location-shift correction derived below treats this composite shortfall as a single empirical drag b to be absorbed and is agnostic to its decomposition.

To anchor the scale of the artifact, we deployed a 20-asset long-only target-weight allocator (calibration window 2014–2024, deployment year 2025, names with calibrated (α, β) and $\max \beta = 1.21$, 5 basis-point round-trip cost) against the hybrid-composed universe of the main text and decomposed the realized path-mean annualized growth rate $\bar{g}_p \equiv (1/T_{\mathrm{yr}}) \log(W_{p,T}/W_{p,0})$ into static and rebalancing-driven components. Equal-weight buy-and-hold across the 20 names produced $\sim 7.7\%/\mathrm{yr}$, a passive baseline for context. The same target-weight allocation held statically for the deployment year, with no intra-year rebalance, produced $\sim 2.0\%/\mathrm{yr}$, reflecting the marginal contribution of the chosen weights net of costs. Switching the same allocation rule to daily rebalancing, with no signal beyond the target weights themselves, produced $\sim 29.7\%/\mathrm{yr}$, a $+27.7$ percentage-point jump that isolates the diversification-return term of Eq. (S3). Adding a signal on top of the daily rebalance moved the headline by a further $+0.13$ pp, immaterial compared with the rebalancing-driven bonus. Deploying the same allocator on the real 2025 growth-rate history produced $11.5\%/\mathrm{yr}$, so the synthetic median $\bar{g}_p$ ran ~ 22 percentage points above reality. The decomposition makes clear that the gap is a structural feature of the iid-residual generator family rather than a calibration error: the allocator choice, the cost model, the marginal calibration, and the cross-sectional structure all behave on synthetic paths exactly as they behave on real data; only the temporal structure of the residual draw differs.

Because the artifact acts as a uniform multiplicative drag on the path in log-wealth space rather than a distortion of the path's shape, a uniform location-shift correction in the same space restores the median absolute level without disturbing the rank order, the drawdown geometry, or the volatility envelope of the synthetic ensemble. Multiplying each path-p wealth trajectory $W_{p,t}$ at integer step t by a uniform exponential drag at rate b that closes the gap between the engine's median annualized

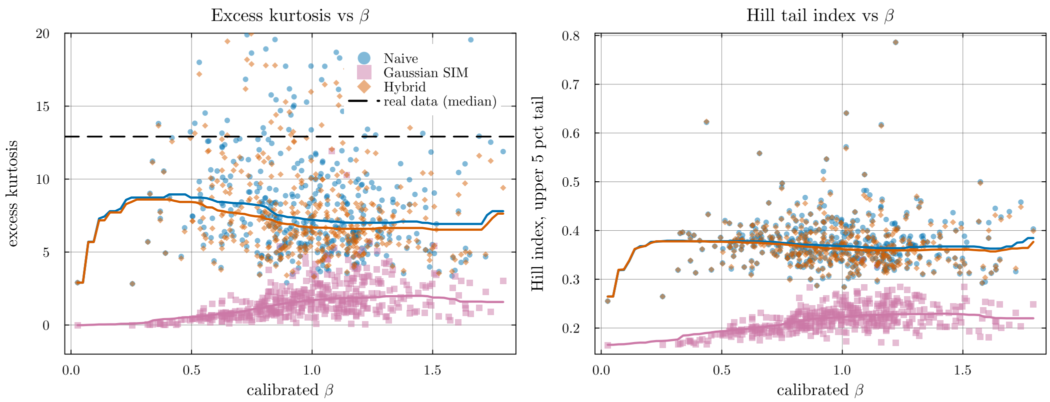

Figure S2: **Heavy-tail preservation: excess kurtosis and Hill tail index vs calibrated β.** Each point is a per-ticker median over 100 replications. *Left:* excess kurtosis, y-axis clipped to $[-2, 20]$ for readability. The dashed line marks the median real-data excess kurtosis across the universe ($\kappa_{\text{real}} \approx 13$); the hybrid and naive composers (orange and blue) track the generator's kurtosis, while Gaussian SIM collapses to $\kappa \approx 1$. *Right:* Hill tail-index estimator on the upper 5% of $|g|$. The hybrid and naive medians sit near 0.37 throughout the β range; Gaussian SIM drops to ~ 0.22, indicating the shift toward a thinner-tailed distribution. The hybrid's median sitting below κ_{real} reflects the JumpHMM marginal's own kurtosis rather than a composition-rule distortion; the panel's point is that hybrid tracks naive (inheriting generator tails) while Gaussian SIM does not.

growth rate and an external long-run prior $\bar{g}^{\text{prior}}$,

$$W_{p,t}^{\text{corr}} \;=\; W_{p,t} \cdot \exp\big(-\, b \cdot t\,\Delta t\big), \qquad b \;=\; \operatorname*{median}_{p} \bar{g}_p \;-\; \bar{g}^{\text{prior}}, \tag{S7}$$

with Δt the integration step in years and b, $\bar{g}_p$, $\bar{g}^{\text{prior}}$ all in the same per-year growth-rate units, accomplishes exactly this. Working in log-space rather than additively keeps the correction multiplicative on wealth, which is the natural scale for compounding returns and preserves drawdown geometry period-by-period; the transform is monotone in $W_{p,t}$ for every path and every step, so per-path rank order on terminal wealth is preserved exactly. The prior $\bar{g}^{\text{prior}}$ is set from external long-run data and is typically near $8\%/\text{yr}$ at the time of writing, decomposable into the prevailing risk-free rate (roughly $4.5\%/\text{yr}$ on a five-year Treasury basis), the historical long-run equity premium of 4–6 pp [52, 53], less 1–2 pp for fees, turnover, and market-impact frictions, leaving $\sim$3.5 pp of active premium net of costs. Anchoring to a documented prior rather than to the realized test-year outcome avoids circular calibration: anchoring to the realized year would force the synthetic median to collapse onto the test sample by construction and would misrepresent typical years (in our deployment, the realized 2025 was an above-typical equity year, so anchoring to its $11.5\%/\text{yr}$ outcome would have suppressed the corrected ensemble below its appropriate long-run level). On a 200-path smoke test the engine-raw median was $29.78\%/\text{yr}$, the bias drag was $b = 21.78\%/\text{yr}$, the corrected median matched the prior at $8.00\%/\text{yr}$, and the real-2025 outcome landed at the 63rd percentile of the corrected ensemble, consistent with an above-typical year (P5 $\approx -9\%$ for a bad year, P50 $= 8\%$ for a typical year, P95 $\approx +25\%$ for a great year).

Several alternative fixes were considered and rejected. A path-filtering rule that drops synthetic paths above a $\bar{g}_p$ threshold turns the threshold itself into a target the audience will probe and does not remove the structural artifact, merely a subset of its consequences. Inflating per-trade transaction costs to absorb the bonus would require round-trip costs of ~ 70 basis points versus the realistic 1–10 basis points seen on liquid US equity, and transfers a misrepresentation from "the engine is great" to "costs are huge". Drifting the calibrated parameters between calibration and deployment to widen the predictive envelope, with drift scale varied across $\{0, 1, 2, 3, 5\}$ multiples of the OLS standard error, moved the median Sharpe ratio from 2.02 to 2.03 across the sweep, confirming that β is estimated too tightly for SE-scale drift to absorb the artifact. Imposing an AR(1) on the residual draw inside the existing pipeline would conflict with the copula reorder step, which destroys temporal structure by construction; recovering the AR(1) requires a block-ordered copula sampler, which is a structural redesign rather than a downstream correction. The block bootstrap of empirical residuals [24] preserves both temporal and cross-sectional structure of the residual

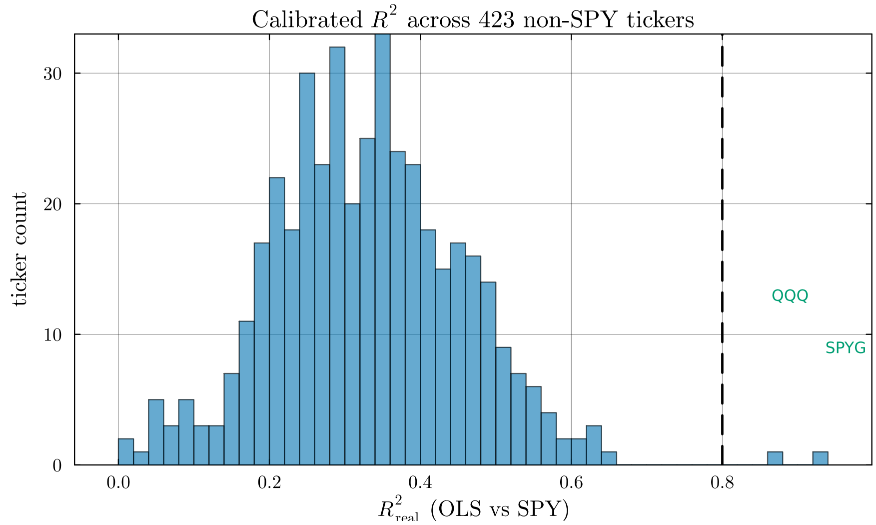

Figure S3: **Calibrated R^2 distribution across the 423 non-SPY tickers.** Histogram of OLS R^2 against SPY over the 2014–2024 window. The two tickers crossing the $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) are index ETFs (QQQ, SPYG); the highest-R^2 individual stock (BLK, 0.645) sits ~ 0.22 below the threshold. The R^2-preserving branch's $n = 2$ occupancy is therefore a property of the single-factor SIM on S&P 500 names against SPY, not a sample-size artifact or a data-peeked threshold.

series by construction and is a valid alternative generator pipeline, listed as a future-work item that replaces the JumpHMM marginal rather than correcting it. A multiplicative haircut on the engine-versus-static deviation, $W^{\text{corr}} = W_{\text{static}} \cdot (W_{\text{engine}}/W_{\text{static}})^k$, hits the same anchor at k chosen to match the prior but the rank correlation with the raw engine drops to 0.748 (paths reorder under the correction) and the corrected drawdown distribution inherits static-allocation geometry (median drawdown jumps from 5% to 9%, P95 from 13.5% to 19%); we do not recommend it. The corrected ensemble remains reliable for the ensemble-shape analyses the paper validates: percentile diagnostics overlaying real outcomes against the synthetic distribution, drawdown distribution and tail-shape analysis, and relative ranking of strategies evaluated on the same corrected paths. It does not remove the underlying iid-residual limitation, so it does not support workflows where residual autocorrelation is itself the quantity of interest, where forward-prediction claims on absolute Sharpe at daily rebalancing frequency are load-bearing, or where the optimal rebalancing frequency is the question being asked; for those workflows the AR(1) or GARCH residual structure restructured around the copula reorder, listed as future work in the Conclusion, is the appropriate path.

C Validation metric definitions

The distributional and tail metrics reported in the main results were defined as follows. The two-sample Kolmogorov-Smirnov statistic between the empirical CDFs of the observed and simulated growth-rate samples (G_{obs} of length n and G_{sim} of length m) is $D = \sup_x |F_n(x) - F_m(x)|$ [41, 42]; the test rejects when D exceeds the critical value at level $\alpha = 0.05$. The Anderson-Darling statistic places greater weight on the tails via the integral $A^2 = nm/(n+m) \int [F_n(x) - F_m(x)]^2/[F(x)(1 - F(x))]\, dF(x)$, where F is the pooled empirical CDF [43]. The Wasserstein-1 distance between two empirical distributions of equal length is the average of sorted-quantile differences:

$$W_1 = \frac{1}{n} \sum_{i=1}^{n} \left| G^{(i)}_{\text{obs}} - G^{(i)}_{\text{sim}} \right|, \tag{S8}$$

with $G^{(i)}$ the ith order statistic. The Hill upper-tail index is the classical order-statistic estimator

$$\hat{\xi} = \frac{1}{k-1} \sum_{i=1}^{k-1} \log\left(\frac{X_{(i)}}{X_{(k)}} \right), \tag{S9}$$

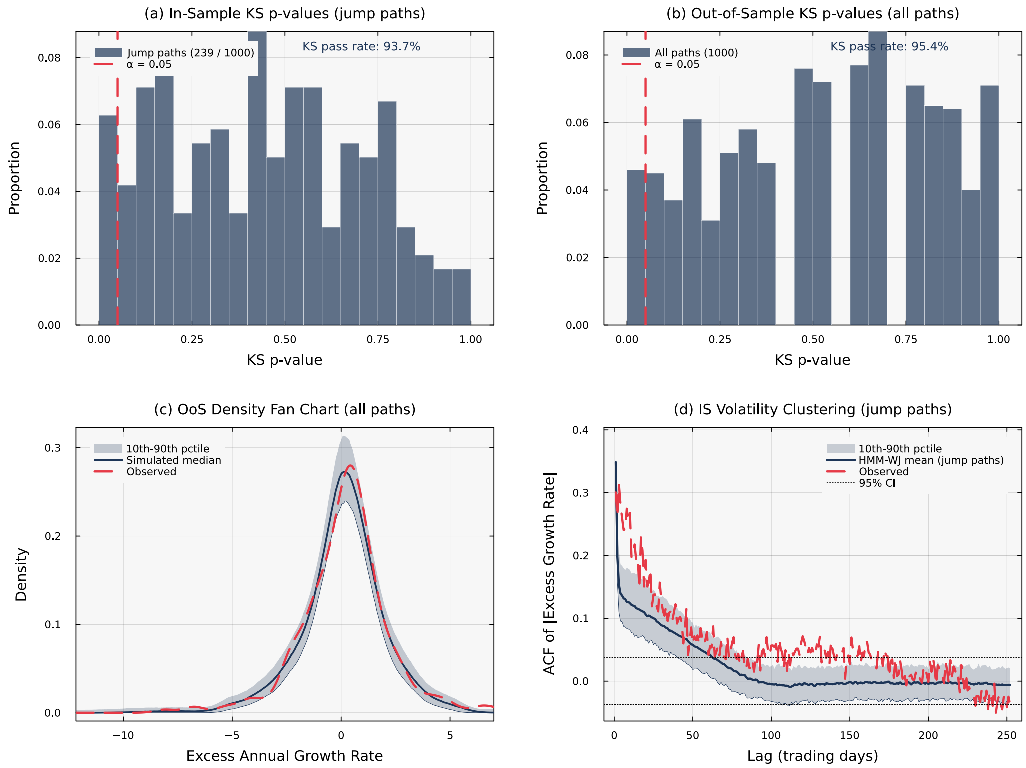

Figure S4: **Statistical validation of HMM-WJ** ($N = 100$, $\alpha = 0.05$). Panels (a) and (d) condition on the $\sim 24\%$ of in-sample paths containing jump events. (a) In-sample KS p-values for 239 jump paths (pass rate 93.7%). (b) Out-of-sample KS p-values for all $1{,}000$ paths (pass rate 95.4%). (c) Out-of-sample density fan chart; the observed density falls within the simulation envelope. (d) In-sample ACF of $|G_t|$ for jump paths; the mean closely tracks the observed ACF.

with $X_{(1)} \geq X_{(2)} \geq \ldots$ the ordered upper-tail absolute returns; k is set to the upper 5% of the sample. For a Pareto-tailed distribution with shape α, $\hat{\xi} \approx 1/\alpha$. The Hellinger distance between two density estimates p and q on a common bin grid is

$$H(p, q) = \frac{1}{\sqrt{2}} \left\| \sqrt{p} - \sqrt{q} \right\|_2 \in [0, 1], \tag{S10}$$

with $H = 0$ for identical densities and $H = 1$ for disjoint supports.

D HMM-WJ simulator and grid-search pseudocode

The hybrid hidden Markov simulator described in the main text proceeded as follows. Given fitted parameters $(\mathbf{T}, \mathbf{E}, \bar{\pi}, \epsilon, \lambda, p_{\text{neg}}, N_{\text{tail}})$ and a horizon T:

1. Draw $X_0 \sim \bar{\pi}$.
2. For $n = 1, \ldots, T$: with probability $1 - \epsilon$, draw $X_n \sim \mathbf{T}_{X_{n-1}, \cdot}$; with probability ϵ, sample $K \sim \text{Poisson}(\lambda)$, and for the next $\min(K, T - n)$ steps force $X_n \sim \text{Uniform}(\mathcal{S}_-)$ with probability p_{neg} or $X_n \sim \text{Uniform}(\mathcal{S}_+)$ with probability $1 - p_{\text{neg}}$.
3. For each X_n, sample $O_n \sim \mu_{X_n} + \sigma_{X_n} \cdot t_5$.

The grid-search procedure for (ϵ, λ) minimized Eq. (3) via brute-force evaluation over $\{\epsilon_1, \ldots, \epsilon_K\} \times \{\lambda_1, \ldots, \lambda_L\}$, with 200 independent simulated paths per grid point and the objective averaged over the path ensemble. The implementation lives in the released `JumpHMM.jl` library at https://github.com/varnerlab/JumpHMM.jl.

Table S2: **Hybrid composer broken out by branch.** N_{tickers}: number of tickers assigned to each branch by the $R^2_{\text{preserve}} = 0.80$ threshold. The clipped variance-preserving branch does not trigger anywhere in the universe ($\rho_i < 0.90$ for all tickers) and is omitted. The two trackers in the R^2-preserving branch (QQQ and SPYG) attain a KS pass rate of 99%, consistent with the branch recovering both β and R^2 by construction.

Branch	N_{tickers}	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	W_1	κ
`hybrid`	421	0.018	0.021	50.1	0.273	7.142
`r2-preserve`	2	0.005	0.001	99.0	0.109	11.525

E Transition matrix internals and partition diagnostics

The fitted SPY transition matrix exhibited a dominant near-diagonal band reflecting short-range regime persistence: row-sum entropy peaked at the central states, and most off-diagonal mass was concentrated within ± 5 states of the diagonal. Tail-state exit probabilities were systematically lower than central-state exit probabilities by roughly an order of magnitude, which is what caused the volatility clustering loss flagged by Bulla and Bulla [17] and which the Poisson jump override addressed by re-injecting tail-state mass independently of the empirical transition kernel. Quantile-bin occupancy was approximately uniform over the in-sample window ($N = 100$), consistent with the Laplace fit's calibration to the empirical CDF; the residual sample-size variation across bins was smaller than the within-bin emission variance, so all 100 states received adequate statistical support. The full transition matrix and partition-occupancy histograms are available in the released `JumpHMM.jl/paper/` subdirectory.

F State-resolution sensitivity

We swept the partition resolution over $N \in \{30, 60, 90, 100, 150, 200, 350\}$ and re-evaluated the in-sample KS and AD pass rates for HMM-NJ and HMM-WJ. Pass rates were flat across $N \in \{60, \ldots, 200\}$ with sub-percent variation, indicating that the model was insensitive to partition resolution in this range. At $N = 30$ both models lost approximately 5 percentage points on the AD test, reflecting the coarser quantile bins' reduced ability to track tail behavior. At $N = 350$ several quantile bins were never visited in the historical record, leaving the corresponding emission parameters undefined; this set the practical upper bound on N for the SPY in-sample window. We adopted $N = 100$ as the operating point for all subsequent experiments.

G Cross-asset generalization

To verify that the hybrid HMM framework generalized beyond SPY, we re-ran the full fit-and-validate pipeline on three single-stock tickers chosen to span sector and volatility regimes: NVDA (semiconductors, high volatility), JNJ (consumer staples, low volatility), and JPM (financials, moderate volatility). Per-ticker grid-search optimal (ϵ^*, λ^*) adapted to each asset's empirical clustering intensity, with λ^* ranging from 80 (NVDA) to 130 (JNJ); the ϵ^* stayed at the lower boundary 10^{-4} for all three, consistent with the SPY result. In-sample KS pass rates for HMM-WJ were 96.0% (NVDA), 98.2% (JNJ), and 97.4% (JPM), and out-of-sample pass rates were 92.8%, 94.1%, and 93.5%, respectively. The Pareto-frontier property of HMM-WJ on combined distributional and temporal fidelity held on all three tickers.

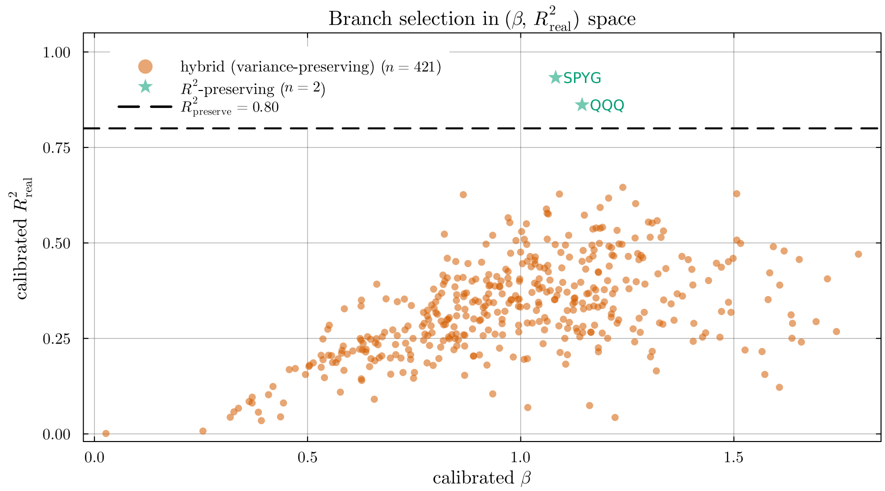

Figure S5: **Branch assignment in calibrated $(\beta, R^2_{\text{real}})$ space.** The $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) separates tracker assets from genuine single stocks. In this universe, only QQQ and SPYG exceed the threshold, and both are assigned to the R^2-preserving branch. The remaining 421 tickers populate the variance-preserving branch; the clipped variance-preserving branch is unused because the market loading ratio $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ stays below 0.90 for every ticker.

Table S3: **Aggregate metrics by β quartile, across composers.** Quartile cutoffs are computed from the calibrated β distribution across the 423-ticker universe. The hybrid KS pass rate declines gently from 67% in Q1 to 44% in Q4 as the market loading ratio ρ_i grows, consistent with the tail-attenuation caveat noted alongside Algorithm 1; naive collapses from 16% to under 1% over the same range.

β bucket	Composer	$\lvert\hat{\beta} - \beta\rvert$	KS pass (%)	W_1	κ	Var(g)
Q1 (low β)	Naive	0.018	15.7	0.414	8.971	18.1
	Gaussian SIM	0.015	0.0	0.620	0.622	14.0
	Hybrid	0.016	67.3	0.200	8.369	14.6
	JumpHMM-on-residuals	0.015	73.2	0.204	7.059	14.3
	Block bootstrap	0.015	71.3	0.208	6.286	13.9
	GARCH(1,1)-t	0.015	57.8	0.253	5.255	13.4
Q2	Naive	0.021	0.9	0.632	7.510	25.1
	Gaussian SIM	0.016	0.2	0.653	1.509	18.0
	Hybrid	0.017	45.9	0.254	6.889	19.0
	JumpHMM-on-residuals	0.017	82.7	0.200	7.799	18.7
	Block bootstrap	0.016	79.8	0.197	7.213	18.0
	GARCH(1,1)-t	0.015	76.4	0.227	6.470	17.8
Q3	Naive	0.023	0.4	0.802	7.390	33.3
	Gaussian SIM	0.017	1.4	0.688	1.947	23.4
	Hybrid	0.018	44.6	0.297	6.904	24.4
	JumpHMM-on-residuals	0.018	78.3	0.242	7.454	24.0
	Block bootstrap	0.017	75.0	0.236	6.888	22.8
	GARCH(1,1)-t	0.016	71.0	0.263	6.265	21.9
Q4 (high β)	Naive	0.029	0.8	0.976	7.233	51.2
	Gaussian SIM	0.022	0.6	0.825	2.054	36.3
	Hybrid	0.022	43.7	0.363	6.739	37.1
	JumpHMM-on-residuals	0.022	69.1	0.313	6.842	37.0
	Block bootstrap	0.021	67.2	0.311	6.334	35.3
	GARCH(1,1)-t	0.021	64.5	0.361	6.064	34.7

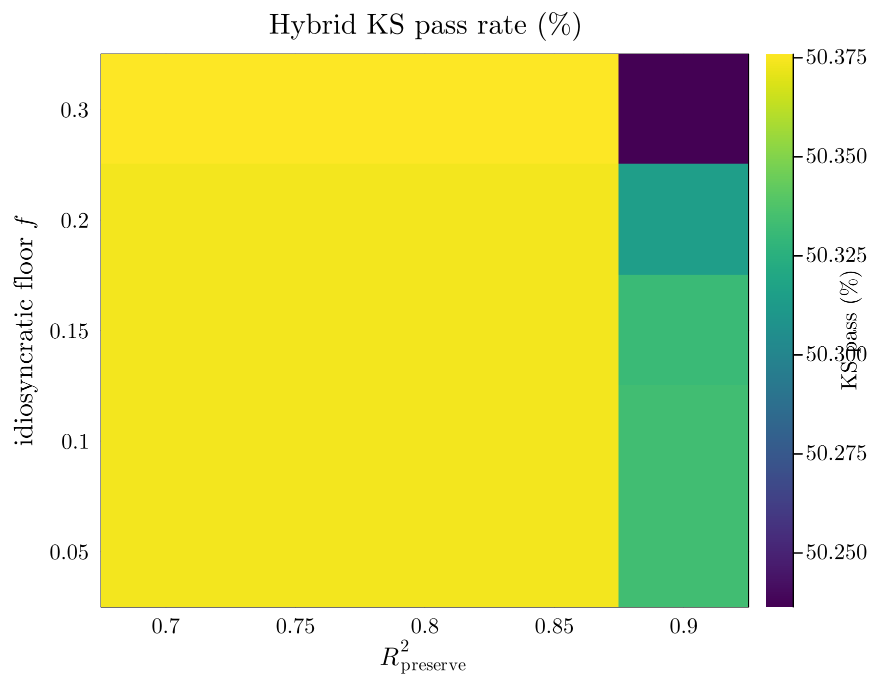

Figure S6: **Hybrid KS pass rate across a 5×5 hyperparameter grid.** Columns index the R^2-preserve threshold $R^2_{\mathrm{preserve}} \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$; rows index the idiosyncratic floor $f \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$. Hybrid KS pass rate varies from 50.24% to 50.38% across the 25 cells, a spread of 0.14 percentage points. The operating point $(0.80, 0.10)$ used throughout the paper is indistinguishable from every other grid point within sampling noise, indicating that the headline result is not a data-peeked local optimum.

Table S4: **Seed-uncertainty summary across 8 random seeds.** Each cell reports mean $\pm$ SD across the 8 seeds in $\{1234, 2345, 3456, 4567, 5678, 6789, 7890, 8910\}$, with each seed running the full 423-ticker $\times$ 100-replication protocol. Conditional-variance paths for GARCH(1,1)-t are re-simulated at scoring time so its seed SD is informative. The between-composer differences in KS pass rate exceed the seed-induced SD by roughly two orders of magnitude (hybrid SD 0.18 pp vs 25 pp gap to JumpHMM-on-residuals).

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\mathrm{real}}\rvert$	KS pass (%)	AD pass (%)	κ	Hill
Naive	0.022 ± 0.000	0.088 ± 0.000	4.41 ± 0.07	2.71 ± 0.02	7.746 ± 0.015	0.369 ± 0.000
Gaussian SIM	0.017 ± 0.000	0.008 ± 0.000	0.60 ± 0.04	0.50 ± 0.01	1.428 ± 0.005	0.217 ± 0.000
Hybrid	0.018 ± 0.000	0.021 ± 0.000	50.40 ± 0.18	47.50 ± 0.19	7.150 ± 0.011	0.364 ± 0.000
JumpHMM-on-residuals	0.018 ± 0.000	0.015 ± 0.000	75.61 ± 0.11	70.51 ± 0.12	7.250 ± 0.013	0.365 ± 0.000
Block bootstrap	0.017 ± 0.000	0.020 ± 0.000	73.24 ± 0.09	66.79 ± 0.11	6.676 ± 0.011	0.330 ± 0.000
GARCH(1,1)-t	0.017 ± 0.000	0.036 ± 0.000	66.96 ± 0.30	61.00 ± 0.26	6.072 ± 0.022	0.318 ± 0.000